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Organometallic compound and organic light-emitting device including the same

Abstract

An organometallic compound is represented by Formula 1. An organic light-emitting device includes: a first electrode; a second electrode facing the first electrode; an organic layer between the first electrode and the second electrode and including an emission layer; and at least one of the organometallic compound represented by Formula 1.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to and the benefit of Korean Patent Application No. 10-2019-0064072, filed on May 30, 2019, in the Korean Intellectual Property Office, the entire content of which is incorporated herein by reference.

BACKGROUND

1. Field

(2) One or more embodiments relate to an organometallic compound and an organic light-emitting device including the same.

2. Description of the Related Art

(3) Organic light-emitting devices are self-emission devices that produce full-color images, and also have wide viewing angles, high contrast ratios, short response times, and excellent characteristics in terms of brightness, driving voltage, and response speed, as compared to other devices in the art.

(4) The organic light-emitting device may include a first electrode on a substrate, and a hole transport region, an emission layer, an electron transport region, and a second electrode, which are sequentially on the first electrode. Holes provided from the first electrode may move toward the emission layer through the hole transport region, and electrons provided from the second electrode may move toward the emission layer through the electron transport region. Carriers, such as holes and electrons, recombine in the emission layer to produce excitons. These excitons transit (e.g., transition or relax) from an excited state to a ground state, thereby generating light.

SUMMARY

(5) One or more embodiments include an organometallic compound and an organic light-emitting device including the same.

(6) Additional aspects of embodiments will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

(7) An aspect of an embodiment of the present disclosure provides an organometallic compound represented by Formula 1:

ML.sub.AL.sub.B. Formula 1

(8) In Formula 1, M may be selected from platinum (Pt), palladium (Pd), copper (Cu), zinc (Zn), silver (Ag), gold (Au), iridium (Ir), rhodium (Rh), cobalt (Co), ruthenium (Ru), rhenium (Re), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), and thulium (Tm), L.sub.A may be a tridentate ligand represented by Formula 1A, and L.sub.B may be a tridentate ligand represented by Formula 1B:

(9) ##STR00001##

(10) In Formulae 1A and 1B, ring A.sub.1 may be a 5-membered ring including Y.sub.1, a C.sub.5-C.sub.60 carbocyclic ring in which a 5-membered ring including Y.sub.1 is condensed, or a C.sub.1-C.sub.60 heterocyclic ring in which a 5-membered ring including Y.sub.1 is condensed, ring A.sub.2 to ring A.sub.6 may each independently be selected from a C.sub.5-C.sub.60 carbocyclic ring and a C.sub.1-C.sub.60 heterocyclic ring, Y.sub.1 to Y.sub.6 may each independently be N or C, L.sub.1 and L.sub.2 may each independently be selected from $*-O-*$, $*-S-*$, $*-C(R.sub.7)(R.sub.8)-*$, $*-C(R.sub.7)=*$, $*=C(R.sub.7)-*$, $*-C(R.sub.7)=C(R.sub.8)-*$, $*-C(=O)-*$, $*-C(=S)-*$, $*-C\equiv C-*$, $*-B(R.sub.7)-*$, $*-N(R.sub.7)-*$, $*-P(R.sub.7)-*$, $*-Si(R.sub.7)(R.sub.8)-*$, $*-P(R.sub.7)(R.sub.8)-*$, and $*-Ge(R.sub.7)(R.sub.8)-*$, n1 may be selected from 1, 2, and 3, and n2 may be selected from 0, 1, 2, and 3, R.sub.1 to R.sub.8 may each independently be selected from hydrogen, deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, $-Si(Q.sub.1)(Q.sub.2)(Q.sub.3)$, $-N(Q.sub.1)(Q.sub.2)$, $-B(Q.sub.1)(Q.sub.2)$, $-C(=O)(Q.sub.1)$, $-S(=O).sub.2(Q.sub.1)$, and $-P(=O)(Q.sub.1)(Q.sub.2)$, and two neighboring R.sub.1(s) among R.sub.1(s) in the number of b1, two neighboring R.sub.2(s) among R.sub.2(s) in the number of b2, two neighboring R.sub.3(s) among R.sub.3(s) in the number of b3, two neighboring R.sub.4(s) among R.sub.4(s) in the number of b4, two neighboring R.sub.5(s) among R.sub.5(s) in the number of b5, two neighboring R.sub.6(s) among R.sub.6(s) in the number of b6, and at least one of R.sub.1 and R.sub.7, R.sub.2 and R.sub.7, and R.sub.3 and R.sub.7 may optionally be linked to form a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, b1 to b6 may each independently be an integer from 1 to 10, * indicates a binding site to M, at least one substituent of the substituted C.sub.5-C.sub.60 carbocyclic group, the substituted C.sub.1-C.sub.60 heterocyclic group, the substituted C.sub.1-C.sub.60 alkyl group, the substituted C.sub.2-C.sub.60 alkenyl group, the substituted C.sub.2-C.sub.60 alkynyl group, the substituted C.sub.1-C.sub.60 alkoxy group, the substituted C.sub.3-C.sub.10 cycloalkyl group, the substituted C.sub.1-C.sub.10 heterocycloalkyl group, the substituted C.sub.3-C.sub.10 cycloalkenyl group, the substituted C.sub.1-C.sub.10 heterocycloalkenyl group, the substituted C.sub.6-C.sub.60 aryl group, the substituted C.sub.6-C.sub.60 aryloxy group, the substituted C.sub.6-C.sub.60 arylthio group, the substituted C.sub.1-C.sub.60 heteroaryl group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group may be selected from: deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-

C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, and a C.sub.1-C.sub.60 alkoxy group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, and a C.sub.1-C.sub.60 alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.11)(C.sub.212)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(=O)(Q.sub.11), —S(=O).sub.2(Q.sub.11), and —P(=O)(Q.sub.11)(Q.sub.12); a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group; a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(=O)(Q.sub.21), —S(=O).sub.2(Q.sub.21), and —P(=O)(Q.sub.21)(Q.sub.22); and —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a biphenyl group, and a terphenyl group.

(11) Another aspect of an embodiment of the present disclosure provides an organic light-emitting device including: a first electrode; a second electrode facing the first electrode; an organic layer between the first electrode and the second electrode and including an emission layer; and at least one organometallic compound represented by Formula 1.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and/or other aspects of embodiments will become apparent and more readily appreciated from the following description of the embodiments, taken in conjunction with the accompanying drawings in which:
- (2) FIG. 1 is a schematic view of an organic light-emitting device according to an embodiment;
- (3) FIG. 2 is a schematic view of an organic light-emitting device according to another embodiment;
- (4) FIG. 3 is a schematic view of an organic light-emitting device according to another embodiment; and
- (5) FIG. 4 is a schematic view of an organic light-emitting device according to another embodiment.

DETAILED DESCRIPTION

(6) Reference will now be made in more detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described herein below, by referring to the figures, to explain aspects of embodiments of the present description. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

(7) An organometallic compound according to an embodiment may be represented by Formula 1:
ML.sub.AL.sub.B. Formula 1

(8) In Formula 1, M may be selected from platinum (Pt), palladium (Pd), copper (Cu), zinc (Zn), silver (Ag), gold (Au), iridium (Ir), rhodium (Rh), cobalt (Co), ruthenium (Ru), rhenium (Re), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), and thulium (Tm).

(9) In one embodiment, M may be iridium or rhodium.

(10) In Formula 1, L.sub.A may be a tridentate ligand represented by Formula 1A, and L.sub.B may be a tridentate ligand represented by Formula 1B:

(11) ##STR00002##

(12) In Formula 1A, ring A.sub.1 may be a 5-membered ring including Y.sub.1, a C.sub.5-C.sub.60 carbocyclic ring in which the 5-membered ring including Y.sub.1 is condensed, or a C.sub.1-C.sub.60 heterocyclic ring in which the 5-membered ring including Y.sub.1 is condensed.

(13) The 5-membered ring including Y.sub.1 refers to a 5-membered ring including Y.sub.1 as a ring-forming atom. The C.sub.5-C.sub.60 carbocyclic ring in which the 5-membered ring including Y.sub.1 is condensed refers to a polycyclic group in which the 5-membered ring including Y.sub.1 and at least one C.sub.5-C.sub.60 carbocyclic ring are condensed. The C.sub.1-C.sub.60 heterocyclic ring in which the 5-membered ring including Y.sub.1 is condensed refers to a polycyclic group in which the 5-membered ring including Y.sub.1 and at least one C.sub.1-C.sub.60 heterocyclic ring are condensed.

(14) ##STR00003##

(15) In Formula 1A, a group represented by may be selected from groups represented by Formulae A1-1 to A1-7:

(16) ##STR00004##

(17) In Formulae A1-1 to A1-7, X.sub.11 may be C(R.sub.11) or N, X.sub.12 may be C(R.sub.12) or N, X.sub.13 may be C(R.sub.13) or N, X.sub.14 may be C(R.sub.14) or N, X.sub.15 may be C(R.sub.15) or N, X.sub.16 may be C(R.sub.16) or N, and X.sub.17 may be C(R.sub.17) or N, X.sub.18 may be C(R.sub.18)(R.sub.19), Si(R.sub.18)(R.sub.19), N(C.sub.18), P(C.sub.18), O, or S, R.sub.11 to R.sub.19 may each independently be the same as defined in connection with R.sub.1,

and * indicates a binding site to M, and *' indicates a binding site to (L.sub.1).sub.a1.

(18) In one embodiment, in Formulae A1-1 to A1-7, R.sub.11 to R.sub.19 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkyl group substituted with at least one —F, a C.sub.1-C.sub.10 alkyl group substituted with at least one —Cl, a C.sub.1-C.sub.10 alkyl group substituted with at least one —Br, a C.sub.1-C.sub.10 alkyl group substituted with at least one —I, a C.sub.1-C.sub.10 alkoxy group, a C.sub.6-C.sub.20 aryl group, and a C.sub.1-C.sub.20 heteroaryl group.

(19) In one embodiment, a group represented by

(20) ##STR00005##

in Formula 1A may be represented by Formula A1-1, X.sub.11 may be C(R.sub.11), X.sub.12 may be C(R.sub.12), and X.sub.13 may be N.

(21) In one embodiment, a group represented by

(22) ##STR00006##

in Formula 1A may be represented by Formula A1-1, X.sub.11 may be N, X.sub.12 may be C(R.sub.12), and X.sub.13 may be N.

(23) In one embodiment, a group represented by

(24) ##STR00007##

in Formula 1A may be represented by Formula A1-1, X.sub.11 may be C(R.sub.11), X.sub.12 may be C(R.sub.12), X.sub.13 may be N, and at least one of Ru and R.sub.12 may be selected from a cyano group, a C.sub.1-C.sub.10 alkyl group substituted with at least one —F, a C.sub.1-C.sub.10 alkyl group substituted with at least one —Cl, a C.sub.1-C.sub.10 alkyl group substituted with at least one —Br, and a C.sub.1-C.sub.10 alkyl group substituted with at least one —I. For example, at least one of Ru and R.sub.12 may be

(25) ##STR00008##

(26) a cyano group or —CF.sub.3. In another embodiment, a group represented by in Formula 1A may be represented by Formula A1-1, X.sub.11 may be N, X.sub.12 may be C(R.sub.12), X.sub.13 may be N, and R.sub.12 may be selected from a cyano group, a C.sub.1-C.sub.10 alkyl group substituted with at least one —F, a C.sub.1-C.sub.10 alkyl group substituted with at least one —Cl, a C.sub.1-C.sub.10 alkyl group substituted with at least one —Br, and a C.sub.1-C.sub.10 alkyl group substituted with at least one —I. For example, R.sub.12 may be a cyano group or —CF.sub.3.

(27) In Formulae 1A and 1B, ring A.sub.2 to ring A.sub.6 may each independently be selected from a C.sub.5-C.sub.60 carbocyclic ring and a C.sub.1-C.sub.60 heterocyclic ring.

(28) In one embodiment, in Formulae 1A and 1B, ring A.sub.3, ring A.sub.4, and ring A.sub.6 may each independently be selected from groups represented by Formulae 2-1(1) to 2-1(27), and ring A.sub.2 and ring A.sub.5 may each independently be selected from groups represented by Formulae 2-2(1) to 2-2(19):

(29) ##STR00009## ##STR00010## ##STR00011## ##STR00012## ##STR00013##
##STR00014## ##STR00015##

(30) In Formulae 2-1(1) to 2-1(27) and 2-2(1) to 2-2(19), Y.sub.15 may be N or C bonded to M, X.sub.21 may be N or C(R.sub.21), X.sub.22 may be N or C(R.sub.22), X.sub.23 may be N or C(R.sub.23), X.sub.24 may be N or C(R.sub.24), X.sub.25 may be N or C(R.sub.25), X.sub.26 may be N or C(R.sub.26), X.sub.27 may be N or C(R.sub.27), and X.sub.28 may be N or C(R.sub.28), X.sub.29 may be C(R.sub.29a)(R.sub.29b), Si(R.sub.29a)(R.sub.29b), N(R.sub.29), O, or S, X.sub.30 may be C(R.sub.30a)(R.sub.30b), Si(R.sub.30a)(R.sub.30b), N(R.sub.30), O, or S, Z.sub.21 and Z.sub.22 may each independently be N or P, R.sub.21 to R.sub.30, R.sub.25a to R.sub.30a and R.sub.25b to R.sub.30b may each independently be the same as defined in connection with R.sub.2 to R.sub.6, * indicates a binding site to M, and *' and *'' each indicate a binding site to a neighboring atom.

(31) In one embodiment, in Formulae 1A and 1B, ring A.sub.2 and ring A.sub.5 may each

independently be a group represented by Formula 2-2(1), A.sub.3 may be a group represented by Formula 2-1(1), and A.sub.4 and A.sub.6 may each independently be a group represented by Formula 2-1(17) or 2-1(26).

(32) In one embodiment, in Formula 1A, ring A.sub.1 may be a group represented by Formula A1-11 or A1-12, ring A.sub.2 may be a group represented by Formula A2-11 or A2-12, and ring A.sub.3 may be a group represented by Formula A3-11, A3-12, or A3-13:

(33) ##STR00016##

(34) In Formulae A1-11, A1-12, A2-11, A2-12, and A3-11 to A3-13, Y.sub.2, Y.sub.3, R.sub.11, R.sub.12, R.sub.2, and R.sub.3 may each independently be the same as described herein, b21 and b32 may each independently be an integer from 1 to 3, b22 may be 1 or 2, b31 may be an integer from 1 to 4, * indicates a binding site to M, *' indicates a binding site to (L.sub.1).sub.n1, and *'' indicates a binding site to (L.sub.2).sub.n2.

(35) In Formulae 1A and 1B, Y.sub.1 to Y.sub.6 may each independently be N or C.

(36) In one embodiment, in Formula 1A, Y.sub.11 and Y.sub.2 may each be N, and Y.sub.3 may be C, a bond between Y.sub.11 and M may be a covalent bond, a bond between Y.sub.2 and M may be a coordinate bond, and a bond between Y.sub.3 and M may be a covalent bond.

(37) In one or more embodiment, in Formula 1B, Y.sub.4 and Y.sub.6 may each be C, and a bond between Y.sub.4 and M and a bond between Y.sub.6 and M may each be a coordinate bond.

(38) In one or more embodiments, in Formula 1B, Y.sub.4 and Y.sub.6 may each be N, and a bond between Y.sub.4 and M and a bond between Y.sub.6 and M may each be a coordinate bond.

(39) In one or more embodiment, in Formula 1B, Y.sub.4 to Y.sub.6 may each be C, Y.sub.5 may be C, a bond between Y.sub.4 and M and a bond between Y.sub.6 and M may each be a coordinate bond, and a bond between Y.sub.5 and M may be a covalent bond.

(40) In one or more embodiments, in Formula 1B, Y.sub.4 and Y.sub.6 may each be N, Y.sub.5 may be C, a bond between Y.sub.4 and M and a bond between Y.sub.6 and M may each be a coordinate bond, and a bond between Y.sub.5 and M may be a covalent bond.

(41) In Formula 1A, L.sub.1 and L.sub.2 may each independently be selected from *—O—*', *—S—*', *—C(R.sub.7)(R.sub.8)—*', *—C(R.sub.7)=*', *=C(R.sub.7)—*', *—C(R.sub.7)=C(R.sub.8)—*', *—C(=O)—*', *—C(=S)—*', *—C≡C—*', *—B(R.sub.7)—*', *—N(R.sub.7)—*', *—P(R.sub.7)—*', *—Si(R.sub.7)(R.sub.8)—*', *—P(R.sub.7)(R.sub.8)—*', and *—Ge(R.sub.7)(R.sub.8)—*''.

(42) In Formula 1A, n1 may be selected from 1, 2, and 3, and n2 may be selected from 0, 1, 2, and 3. When n2 is 0, a bond represented by *-(L.sub.2).sub.n2-*' may be a single bond, and ring A.sub.2 and ring A.sub.3 may be linked via a single bond.

(43) In one embodiment, n1 may be 1, and n2 may be 0; or n1 may be 1, and n2 may be 1.

(44) In one embodiment, n1 may be 1, n2 may be 0, and L.sub.1 may be selected from *—O—*', *—C(R.sub.7)(R.sub.8)—*', *—B(R.sub.7)—*', *—N(R.sub.7)—*', and *—Si(R.sub.7)(R.sub.8)—*''.

(45) In one or more embodiments, n1 may be 1, n2 may be 1, and L.sub.1 and L.sub.2 may each independently be selected from *—O—*', *—C(R.sub.7)(R.sub.8)—*', *—B(R.sub.7)—*', and *—N(R.sub.7)—*''.

(46) For example, n1 and n2 may each be 1, and L.sub.1 may be *—O—*', and L.sub.2 may be *—O—*', L.sub.1 may be *—C(R.sub.7)(R.sub.8)—*', and L.sub.2 may be *—O—*', L.sub.1 may be *—N(R.sub.7)—*', and L.sub.2 may be *—O—*', L.sub.1 may be *—S—*', and L.sub.2 may be *—O—*', L.sub.1 may be *—B(R.sub.7)—*', and L.sub.2 may be *—O—*', L.sub.1 may be *—Si(R.sub.7)(R.sub.8)—*', and L.sub.2 may be *—O—*', L.sub.1 may be *—O—*', and L.sub.2 may be *—N(R.sub.7)—*', L.sub.1 may be *—C(R.sub.7)(R.sub.8)—*', and L.sub.2 may be *—N(R.sub.7)—*', L.sub.1 may be *—N(R.sub.7)—*', and L.sub.2 may be *—N(R.sub.7)—*', L.sub.1 may be *—S—*', and L.sub.2 may be *—N(R.sub.7)—*', L.sub.1 may be *—B(R.sub.7)—*', and L.sub.2 may be *—N(R.sub.7)—*', or L.sub.1 may be *—Si(R.sub.7)(R.sub.8)—*', and

L.sub.2 may be *—N(R.sub.7)—*.

(47) In one embodiment, in Formula 1, L.sub.A may be a group represented by Formula 1C:

(48) ##STR00017##

(49) In Formula 1C, ring A.sub.1, ring A.sub.2, Y.sub.1 to Y.sub.3, L.sub.1, n1, R.sub.1, R.sub.2, b1, and b2 may each independently be the same as described herein, X.sub.31 may be C(R.sub.31) or N, X.sub.32 may be C(R.sub.32) or N, X.sub.33 may be C(R.sub.33) or N, X.sub.34 may be C(R.sub.34) or N, X.sub.35 may be C(R.sub.35) or N, X.sub.36 may be C(R.sub.36) or N, and X.sub.37 may be C(R.sub.37) or N, R.sub.31 to R.sub.37 may each independently be the same as defined in connection with R.sub.3 in Formula 1A, and * indicates a binding site to M.

(50) In one embodiment, in Formula 1C, X.sub.31 may be C(R.sub.31), X.sub.32 may be C(R.sub.32) or N, X.sub.33 may be C(R.sub.33), X.sub.34 may be C(R.sub.34), X.sub.35 may be C(R.sub.35) or N, X.sub.36 may be C(R.sub.36), and X.sub.37 may be C(R.sub.37), R.sub.31, R.sub.33, R.sub.34, R.sub.36, and R.sub.37 may each independently be hydrogen, R.sub.32 and R.sub.35 may each independently be selected from hydrogen, a cyano group, a methyl group, —CF.sub.3, —CCl.sub.3, —CBr.sub.3, —Cl.sub.3, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, and a phenyl group.

(51) In one or more embodiments, in Formula 1C, X.sub.31 may be C(R.sub.31), X.sub.32 may be C(R.sub.32) or N, X.sub.33 may be C(R.sub.33), X.sub.34 may be C(R.sub.34), X.sub.35 may be C(R.sub.35) or N, X.sub.36 may be C(R.sub.36), and X.sub.37 may be C(R.sub.37), R.sub.31, R.sub.33, R.sub.34, R.sub.36, and R.sub.37 may each independently be hydrogen, and R.sub.32 and R.sub.35 may each independently be selected from a cyano group and —CF.sub.3.

(52) In one embodiment, in Formula 1, L.sub.B may be a group represented by Formula 1B-11 or 1B-12:

(53) ##STR00018##

(54) In Formulae 1B-11 and 1B-12, Y.sub.5, ring A.sub.5, R.sub.5, and b5 may each independently be the same as described herein, Z.sub.41 to Z.sub.44 may each independently be N or P, X.sub.41 may be C(R.sub.41) or N, X.sub.42 may be C(R.sub.42) or N, X.sub.43 may be C(R.sub.43) or N, and X.sub.44 may be C(R.sub.44) or N, R.sub.41 to R.sub.46 may each independently be the same as defined in connection with R.sub.4 and R.sub.6 in Formula 1B, and * indicates a binding site to M.

(55) In one embodiment, in Formulae 1B-11 and 1B-12, Z.sub.41 to Z.sub.44 may each be N.

(56) In one embodiment, in Formula 1, L.sub.B may be a group represented by Formula 1B-21 or 1B-22:

(57) ##STR00019##

(58) In Formulae 1B-21 and 1B-22, Y.sub.5, X.sub.41 to X.sub.44, and R.sub.41 to R.sub.46 may each independently be the same as described herein, X.sub.51 may be C(R.sub.51) or N, X.sub.52 may be C(R.sub.52) or N, and X.sub.53 may be C(R.sub.53) or N, R.sub.51 to R.sub.53 may each independently be the same as defined in connection with R.sub.5 in Formula 1B, and * indicates a binding site to M.

(59) In one embodiment, in Formulae 1B-21 and 1B-22, X.sub.51 may be C(R.sub.51), X.sub.52 may be C(R.sub.52), and X.sub.53 may be C(R.sub.53), and R.sub.51 to R.sub.53 may each independently be selected from hydrogen, a cyano group, a methyl group, —CF.sub.3, —CCl.sub.3, —CBr.sub.3, —Cl.sub.3, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, and a phenyl group.

(60) In Formulae 1A and 1B, R.sub.1 to R.sub.8 may each independently be selected from hydrogen, deuterium, —F —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60

alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(=O)(Q.sub.1), —S(=O).sub.2(Q.sub.1), and —P(=O)(Q.sub.1)(Q.sub.2).

(61) Two neighboring R.sub.1(s) among R.sub.1(s) in the number of b1, two neighboring R.sub.2(s) among R.sub.2(s) in the number of b2, two neighboring R.sub.3(s) among R.sub.3(s) in the number of b3, two neighboring R.sub.4(s) among R.sub.4(s) in the number of b4, two neighboring R.sub.5(s) among R.sub.5(s) in the number of b5, two neighboring R.sub.6(s) among R.sub.6(s) in the number of b6, and at least one of R.sub.1 and R.sub.7, R.sub.2 and R.sub.7, and R.sub.3 and R.sub.7 may optionally be linked to form a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group.

(62) In one embodiment, R.sub.1 to R.sub.8 in Formulae 1A and 1B may each independently be selected from: hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, and a C.sub.1-C.sub.20 alkoxy group; a C.sub.1-C.sub.20 alkyl group and a C.sub.1-C.sub.20 alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, and a biphenyl group; a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a spiro-fluorene-benzofluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a pyrenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, a silolyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an indolyl group, an isoindolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzosilolyl group, a benzothiazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a dibenzosilolyl group, a carbazolyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a thiadiazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an oxazolopyridinyl group, a thiazolopyridinyl group, a benzonaphthyridinyl group, an azafluorenyl group, an azaspiro-bifluorenyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azadibenzosilolyl group, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), and —B(Q.sub.1)(Q.sub.2); and a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a spiro-fluorene-benzofluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a pyrenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, a silolyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group,

a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an indolyl group, an isoindolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzosilolyl group, a benzothiazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a dibenzosilolyl group, a carbazolyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a thiadiazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an oxazolopyridinyl group, a thiazolopyridinyl group, a benzonaphthyridinyl group, an azafluorenyl group, an azaspiro-bifluorenyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, and an azadibenzosilolyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a spiro-fluorene-benzofluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a pyrenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, a silolyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an indolyl group, an isoindolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzosilolyl group, a benzothiazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a dibenzosilolyl group, a carbazolyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a thiadiazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an oxazolopyridinyl group, a thiazolopyridinyl group, a benzonaphthyridinyl group, an azafluorenyl group, an azaspiro-bifluorenyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azadibenzosilolyl group, a biphenyl group, and a terphenyl group.

(63) In one or more embodiments, R.sub.1 to R.sub.8 in Formulae 1A and 1B may each independently be selected from: hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, and a C.sub.1-C.sub.20 alkoxy group; a C.sub.1-C.sub.20 alkyl group and a C.sub.1-C.sub.20 alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, and a biphenyl group; a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, and a triazinyl group; and a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, and a triazinyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, —CF.sub.3, —CCl.sub.3, —CBr.sub.3, —Cl.sub.3, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, and a triazinyl group.

(64) In one or more embodiments, R.sub.1 to R.sub.8 in Formulae 1A and 1B may each independently be selected from: hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a methyl group, —CF.sub.3, —CCl.sub.3, —CBr.sub.3, —Cl.sub.3, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a sec-butyl group, and a tert-butyl group; a phenyl group and a pyridinyl group; and a phenyl group and a pyridinyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a cyano group, a methyl group, —CF.sub.3, —CCl.sub.3, —CBr.sub.3, —Cl.sub.3, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group.

(65) In one embodiment, at least one selected from R.sub.1(s) in the number of b1, R.sub.2(s) in the number of b2, R.sub.3(s) in the number of b3, R.sub.4(s) in the number of b4, R.sub.5(s) in the number of b5, and R.sub.6(s) in the number of b6 may be an electron withdrawing group, For example, at least one selected from R.sub.1(s) in the number of b1, R.sub.2(s) in the number of b2, R.sub.3(s) in the number of b3, R.sub.4(s) in the number of b4, R.sub.5(s) in the number of b5, and R.sub.6(s) in the number of b6 may be selected from a cyano group, a C.sub.1-C.sub.10 alkyl group substituted with at least one —F, a C.sub.1-C.sub.10 alkyl group substituted with at least one —Cl, a C.sub.1-C.sub.10 alkyl group substituted with at least one —Br, and a C.sub.1-C.sub.10 alkyl group substituted with at least one —I.

(66) In one or more embodiments, at least one selected from R.sub.1(s) in the number of b1, R.sub.2(s) in the number of b2, R.sub.3(s) in the number of b3, R.sub.4(s) in the number of b4, R.sub.5(s) in the number of b5, and R.sub.6(s) in the number of b6 may be selected from a cyano group, —CF.sub.3, —CCl.sub.3, —CBr.sub.3, and —Cl.sub.3. For example, at least one selected from R.sub.1(s) in the number of b1, R.sub.2(s) in the number of b2, R.sub.3(s) in the number of b3, R.sub.4(s) in the number of b4, R.sub.5(s) in the number of b5, and R.sub.6(s) in the number of b6 may be selected from a cyano group and —CF.sub.3.

(67) When the organometallic compound represented by Formula 1 includes at least one electron withdrawing group such as a cyano group, an emission wavelength of the organometallic compound may be shifted to a shorter wavelength (blue shift), and thus, blue light having high color purity is emitted.

(68) In Formulae 1A and 1B, b1 to b6 may each independently be an integer from 1 to 10.

(69) In one embodiment, the organometallic compound may be selected from Compounds 1 to 66:

(70) ##STR00020## ##STR00021## ##STR00022## ##STR00023## ##STR00024##
##STR00025## ##STR00026## ##STR00027## ##STR00028## ##STR00029## ##STR00030##
##STR00031## ##STR00032## ##STR00033## ##STR00034## ##STR00035## ##STR00036##
##STR00037## ##STR00038## ##STR00039## ##STR00040## ##STR00041##

(71) When ring A.sub.2 in Formula 1A is a pyridine ring in which Y.sub.2 is N, most of the lowest unoccupied molecular orbital (LUMO) of ring A.sub.2 (or Formula 1A) is distributed in the pyridine moiety. Because the organometallic compound represented by Formula 1 includes the tridentate ligand L.sub.A in which ring A.sub.1 and ring A.sub.3 are covalently bonded on (or to) both sides of ring A.sub.2, a stability of the bond between ring A.sub.2 and the metal center M.sub.1 may increase. For example, a stability of the bond between ring A.sub.2 and the metal center M.sub.1 may be strengthened to increase the energy required for the organometallic compound to transit (e.g., transition) from a triplet state to a metal centered state (.sup.3MC state). In some embodiments, increasing the stability of the bond between ring A.sub.2 and the metal center M.sub.1 increases the energy level of the .sup.3MC state of the organometallic compound. Therefore, the lifespan of the organometallic compound may be improved, and the organic light-emitting device including the organometallic compound may have a long lifespan.

(72) In the organometallic compound represented by Formula 1, because ring A.sub.1 and ring A.sub.2 are linked in the tridentate ligand L.sub.A via a linker other than a single bond, π -conjugation between ring A.sub.1 and ring A.sub.2 is reduced and the adjustment of the LUMO energy level is facilitated. Therefore, the emission wavelength of the organometallic compound

may be shifted to a shorter wavelength such that the organometallic compound is configured to emit blue light having high color purity. Furthermore, the adjustment of the emission wavelength of the organometallic compound may be facilitated by suitable selection of L.sub.1 and L.sub.2 in Formula 1A.

(73) Because the organometallic compound represented by Formula 1 includes the tridentate ligand L.sub.B, a highest occupied molecular orbital (HOMO) energy level is increased to shift an emission wavelength of the organometallic compound to a shorter wavelength (blue shift) such that the organometallic compound is configured to emit blue light having high color purity.

(74) Furthermore, the organometallic compound according to one or more embodiments may include at least one electron withdrawing group in a molecule. Therefore, the emission wavelength of the organometallic compound is shifted to a shorter wavelength (blue shift) such that the organometallic compound is configured to emit blue light having high color purity.

(75) A synthesis method for the organometallic compound represented by Formula 1 would be apparent to those of ordinary skill in the art by referring to the following examples.

(76) The organometallic compound represented by Formula 1 may be used between a pair of electrodes of an organic light-emitting device. For example, the organometallic compound may be included in an emission layer. The organometallic compound may act as a dopant in the emission layer. In one or more embodiments, the organometallic compound of Formula 1 may be used as a material for a capping layer located outside a pair of electrodes of an organic light-emitting device.

(77) Accordingly, another aspect of an embodiment of the present disclosure provides an organic light-emitting device including: a first electrode; a second electrode facing the first electrode; an organic layer between the first electrode and the second electrode and including an emission layer; and at least one organometallic compound represented by Formula 1.

(78) In one embodiment, the organic layer of the organic light-emitting device may include at least one organometallic compound represented by Formula 1.

(79) The expression “(an organic layer) includes at least one organometallic compound,” as used herein, may include a case in which “(an organic layer) includes identical compounds represented by Formula 1” and a case in which “(an organic layer) includes two or more different organometallic compounds represented by Formula 1.”

(80) For example, the organic layer may include, as the organometallic compound, only Compound 1. In this regard, Compound 1 may exist only in the emission layer of the organic light-emitting device. In one or more embodiments, the organic layer may include, as the organometallic compound, Compound 1 and Compound 2. In this regard, Compound 1 and Compound 2 may exist in an identical layer (for example, Compound 1 and Compound 2 may all exist in an emission layer), or different layers (for example, Compound 1 may exist in an emission layer and Compound 2 may exist in an electron transport region).

(81) In one or more embodiments, the emission layer of the organic light-emitting device may include the organometallic compound.

(82) In one or more embodiments, the emission layer of the organic light-emitting device may include the organometallic compound, and may further include a host. Here, an amount of the host included in the emission layer may be greater than that of the organometallic compound included in the emission layer. For example, the amount of the organometallic compound may be in a range of about 0.01 parts by weight to about 30 parts by weight based on 100 parts by weight of the emission layer.

(83) In one embodiment, the emission layer may include the organometallic compound, and may be configured to emit blue light or green light. In one embodiment, the emission layer may include the organometallic compound, and may be configured to emit light having a wavelength of maximum emission of about 450 nm or more and about 550 nm or less. For example, the emission layer may be configured to emit blue light having a maximum emission wavelength of about 460 nm or more and about 480 nm or less.

(84) The wavelength of maximum emission of the organometallic compound is obtained by using a photoluminescence (PL) measuring meter.

(85) In one embodiment, the first electrode may be an anode, and the second electrode may be a cathode, and the organic layer may further include a hole transport region between the first electrode and the emission layer and an electron transport region between the emission layer and the second electrode, and the hole transport region may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof, and the electron transport region may include a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, an electron injection layer, or any combination thereof.

(86) In one embodiment, the hole transport region of the organic light-emitting device may include a p-dopant having a LUMO energy level of about -3.5 eV or less.

(87) The term “organic layer,” as used herein, refers to a single layer and/or a plurality of layers between the first electrode and the second electrode of the organic light-emitting device. A material included in the “organic layer” is not limited to an organic material. For example, the organic layer may include an inorganic material.

(88) Description of FIG. 1

(89) FIG. 1 is a schematic cross-sectional view of an organic light-emitting device **10** according to an embodiment. The organic light-emitting device **10** includes a first electrode **110**, an organic layer **150**, and a second electrode **190**.

(90) Hereinafter, the structure of the organic light-emitting device **10** according to an embodiment and a method of manufacturing the organic light-emitting device **10** will be described in connection with FIG. 1.

(91) First Electrode **110**

(92) In FIG. 1, a substrate may be additionally under the first electrode **110** or above the second electrode **190**. The substrate may be a glass substrate or a plastic substrate, each having excellent mechanical strength, thermal stability, transparency, surface smoothness, ease of handling, and water resistance.

(93) The first electrode **110** may be formed by depositing or sputtering a material for forming the first electrode **110** on the substrate. When the first electrode **110** is an anode, the material for forming the first electrode **110** may be selected from materials with a high work function to facilitate hole injection.

(94) The first electrode **110** may be a reflective electrode, a semi-reflective electrode, or a transmissive electrode. When the first electrode **110** is a transmissive electrode, a material for forming a first electrode may be selected from indium tin oxide (ITO), indium zinc oxide (IZO), tin oxide (SnO₂), zinc oxide (ZnO), and any combinations thereof, but embodiments of the present disclosure are not limited thereto. In one or more embodiments, when the first electrode **110** is a semi-transmissive electrode or a reflectable electrode, a material for forming a first electrode may be selected from magnesium (Mg), silver (Ag), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), and any combinations thereof, but embodiments of the present disclosure are not limited thereto.

(95) The first electrode **110** may have a single-layered structure, or a multi-layered structure including two or more layers. For example, the first electrode **110** may have a three-layered structure of ITO/Ag/ITO, but the structure of the first electrode **110** is not limited thereto.

(96) Organic Layer **150**

(97) The organic layer **150** is on the first electrode **110**. The organic layer **150** may include an emission layer.

(98) The organic layer **150** may further include a hole transport region between the first electrode **110** and the emission layer and an electron transport region between the emission layer and the second electrode **190**.

(99) Hole Transport Region in Organic Layer **150**

(100) The hole transport region may have i) a single-layered structure including a single layer including a single material, ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

(101) The hole transport region may include at least one layer selected from a hole injection layer, a hole transport layer, an emission auxiliary layer, and an electron blocking layer.

(102) For example, the hole transport region may have a single-layered structure including a single layer including a plurality of different materials, or a multi-layered structure having a hole injection layer/hole transport layer structure, a hole injection layer/hole transport layer/emission auxiliary layer structure, a hole injection layer/emission auxiliary layer structure, a hole transport layer/emission auxiliary layer structure, or a hole injection layer/hole transport layer/electron blocking layer structure, wherein for each structure, constituting layers are sequentially stacked from the first electrode **110** in this stated order, but the structure of the hole transport region is not limited thereto.

(103) The hole transport region may include at least one selected from m-MTDATA, TDATA, 2-TNATA, NPB(NPD), β -NPB, TPD, Spiro-TPD, Spiro-NPB, methylated-NPB, TAPC, HMTDP, 4,4',4''-tris(N-carbazolyl)triphenylamine (TCTA), polyaniline/dodecylbenzenesulfonic acid (PANI/DBSA), poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate) (PEDOT/PSS), polyaniline/camphor sulfonic acid (PANI/CSA), polyaniline/poly(4-styrenesulfonate) (PANI/PSS), a compound represented by Formula 201, and a compound represented by Formula 202:

(104) ##STR00042## ##STR00043## ##STR00044## ##STR00045##

(105) In Formulae 201 and 202, L.sub.201 to L.sub.204 may each independently be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group, L.sub.205 may be selected from *—O—*, *—S—*, *—N(Q.sub.201)—*, a substituted or unsubstituted C.sub.1-C.sub.20 alkylene group, a substituted or unsubstituted C.sub.2-C.sub.20 alkenylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group, xa1 to xa4 may each independently be an integer from 0 to 3, xa5 may be an integer from 1 to 10, and R.sub.201 to R.sub.204 and Q.sub.201 may each independently be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

(106) For example, in Formula 202, R.sub.201 and R.sub.202 may optionally be linked to each other via a single bond, a dimethyl-methylene group, or a diphenyl-methylene group, R.sub.203 and R.sub.204 may optionally be linked to each other via a single bond, a dimethyl-methylene group, or a diphenyl-methylene group.

(107) In one embodiment, in Formulae 201 and 202, L.sub.201 to L.sub.205 may each independently be selected from: a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylene group, a heptalenylene group, an indacenylene group, an acenaphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a naphthacenylene group, a picenylene group, a perylenylene group, a pentaphenylene group, a hexacenylene group, a pentacenylene group, a rubicenylene group, a coronenylene group, an ovalenylene group, a thiophenylene group, a furanylene group, a carbazolyene group, an indolyene group, an isoindolyene group, a benzofuranylene group, a benzothiophenylene group, a dibenzofuranylene group, a dibenzothiophenylene group, a benzocarbazolyene group, a dibenzocarbazolyene group, a dibenzosilolyene group, and a pyridinylene group; and a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylene group, a heptalenylene group, an indacenylene group, an acenaphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a naphthacenylene group, a picenylene group, a perylenylene group, a pentaphenylene group, a hexacenylene group, a pentacenylene group, a rubicenylene group, a coronenylene group, an ovalenylene group, a thiophenylene group, a furanylene group, a carbazolyene group, an indolyene group, an isoindolyene group, a benzofuranylene group, a benzothiophenylene group, a dibenzofuranylene group, a dibenzothiophenylene group, a benzocarbazolyene group, a dibenzocarbazolyene group, a dibenzosilolyene group, and a pyridinylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C.sub.1-C.sub.10 alkyl group, a phenyl group substituted with —F, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), and —N(Q.sub.31)(Q.sub.32), and Q.sub.31 to Q.sub.33 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

(108) In one or more embodiments, xa1 to xa4 may each independently be 0, 1, or 2.

(109) In one or more embodiments, xa5 may be 1, 2, 3, or 4.

(110) In one or more embodiments, R.sub.201 to R.sub.204 and Q.sub.201 may each independently be selected from:

(111) a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a

perylene group, a pentaphenyl group, a hexacene group, a pentacene group, a rubicene group, a coronene group, an ovalene group, a thiophene group, a furan group, a carbazole group, an indole group, an isoindole group, a benzofuran group, a benzothiophene group, a dibenzofuran group, a dibenzothiophene group, a benzocarbazole group, a dibenzocarbazole group, a dibenzosilole group, and a pyridine group; and a phenyl group, a biphenyl group, a terphenyl group, a pentalene group, an indene group, a naphthyl group, an azulene group, a heptalene group, an indacene group, an acenaphthyl group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picenyl group, a perylene group, a pentaphenyl group, a hexacene group, a pentacene group, a rubicene group, a coronene group, an ovalene group, a thiophene group, a furan group, a carbazole group, an indole group, an isoindole group, a benzofuran group, a benzothiophene group, a dibenzofuran group, a dibenzothiophene group, a benzocarbazole group, a dibenzocarbazole group, a dibenzosilole group, and a pyridine group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C.sub.1-C.sub.10 alkyl group, a phenyl group substituted with —F, a pentalene group, an indene group, a naphthyl group, an azulene group, a heptalene group, an indacene group, an acenaphthyl group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picenyl group, a perylene group, a pentaphenyl group, a hexacene group, a pentacene group, a rubicene group, a coronene group, an ovalene group, a thiophene group, a furan group, a carbazole group, an indole group, an isoindole group, a benzofuran group, a benzothiophene group, a dibenzofuran group, a dibenzothiophene group, a benzocarbazole group, a dibenzocarbazole group, a dibenzosilole group, a pyridine group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), and —N(Q.sub.31)(Q.sub.32), and Q.sub.31 to Q.sub.33 are the same as described herein above.

(112) In one or more embodiments, at least one selected from R.sub.201 to R.sub.203 in Formulae 201 may each independently be selected from: a fluorene group, a spiro-bifluorene group, a carbazole group, a dibenzofuran group, and a dibenzothiophene group; and a fluorene group, a spiro-bifluorene group, a carbazole group, a dibenzofuran group, and a dibenzothiophene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C.sub.1-C.sub.10 alkyl group, a phenyl group substituted with —F, a naphthyl group, a fluorene group, a spiro-bifluorene group, a carbazole group, a dibenzofuran group, and a dibenzothiophene group; but embodiments of the present disclosure are not limited thereto.

(113) In one or more embodiments, in Formula 202, i) R.sub.201 and R.sub.202 may be linked to each other via a single bond, and/or ii) R.sub.203 and R.sub.204 may be linked to each other via a single bond.

(114) In one or more embodiments, at least one selected from R.sub.201 to R.sub.204 in Formula 202 may be selected from: a carbazole group; and a carbazole group substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a

cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C.sub.1-C.sub.10 alkyl group, a phenyl group substituted with —F, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group, but embodiments of the present disclosure are not limited thereto.

(115) The compound represented by Formula 201 may be represented by Formula 201-1 below, but embodiments of the present disclosure are not limited thereto:

(116) ##STR00046##

(117) In one embodiment, the compound represented by Formula 201 may be represented by Formula 201-2 below, but embodiments of the present disclosure are not limited thereto:

(118) ##STR00047##

(119) In one or more embodiments, the compound represented by Formula 201 may be represented by Formula 201-2(1) below, but embodiments of the present disclosure are not limited thereto:

(120) ##STR00048##

(121) In one or more embodiments, the compound represented by Formula 201 may be represented by Formula 201A, but embodiments of the present disclosure are not limited thereto:

(122) ##STR00049##

(123) In one or more embodiments, the compound represented by Formula 201 may be represented by Formula 201A(1) below, but embodiments of the present disclosure are not limited thereto:

(124) ##STR00050##

(125) In one or more embodiments, the compound represented by Formula 201 may be represented by Formula 201A-1 below, but embodiments of the present disclosure are not limited thereto:

(126) ##STR00051##

(127) In one embodiment, the compound represented by Formula 202 may be represented by Formula 202-1 below, but embodiments of the present disclosure are not limited thereto:

(128) ##STR00052##

(129) In one or more embodiments, the compound represented by Formula 202 may be represented by Formula 202-1(1) below, but embodiments of the present disclosure are not limited thereto:

(130) ##STR00053##

(131) In one or more embodiments, the compound represented by Formula 202 may be represented by Formula 202A below, but embodiments of the present disclosure are not limited thereto:

(132) ##STR00054##

(133) In one or more embodiments, the compound represented by Formula 202 may be represented by Formula 202A-1 below, but embodiments of the present disclosure are not limited thereto:

(134) ##STR00055##

(135) In Formulae 201-1, 201-2, 201-2(1), 201A, 201A(1), 201A-1, 202-1, 202-1(1), 202A, and 202A-1, L.sub.201 to L.sub.203, xa1 to xa3, xa5, and R.sub.202 to R.sub.204 are the same as described herein above, L.sub.205 may be selected from a phenylene group and a fluorenylene group, X.sub.211 may be selected from O, S, and N(R.sub.211), X.sub.212 may be selected from O, S, and N(R.sub.212), R.sub.211 and R.sub.212 may each independently be the same as defined in connection with R.sub.203, and R.sub.213 to R.sub.217 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C.sub.1-C.sub.10 alkyl group, a phenyl group substituted with —F, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl

group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group.

(136) The hole transport region may include at least one compound selected from Compounds HT1 to HT48, but embodiments of the present disclosure are not limited thereto:

(137) ##STR00056## ##STR00057## ##STR00058## ##STR00059## ##STR00060##

##STR00061## ##STR00062## ##STR00063## ##STR00064## ##STR00065## ##STR00066##

##STR00067## ##STR00068## ##STR00069## ##STR00070## ##STR00071##

(138) A thickness of the hole transport region may be in a range of about 100 Å to about 10,000 Å, for example, about 100 Å to about 1,000 Å. When the hole transport region includes at least one selected from a hole injection layer and a hole transport layer, the thickness of the hole injection layer may be in a range of about 100 Å to about 9,000 Å, and for example, about 100 Å to about 1,000 Å, and the thickness of the hole transport layer may be in a range of about 50 Å to about 2,000 Å, and for example, about 100 Å to about 1500 Å. When the thicknesses of the hole transport region, the hole injection layer, and the hole transport layer are within these ranges, suitable or satisfactory hole transporting characteristics may be obtained without a substantial increase in driving voltage.

(139) The emission auxiliary layer may increase light-emission efficiency by compensating for an optical resonance distance according to the wavelength of light emitted by an emission layer, and the electron blocking layer may block the flow of electrons from an electron transport region. The emission auxiliary layer and the electron blocking layer may include the materials as described herein above.

(140) p-Dopant

(141) The hole transport region may further include, in addition to these materials, a charge-generation material for the improvement of conductive properties. The charge-generation material may be homogeneously or non-homogeneously dispersed in the hole transport region.

(142) The charge-generation material may be, for example, a p-dopant.

(143) In one embodiment, the p-dopant may have a LUMO energy level of about -3.5 eV or less.

(144) The p-dopant may include at least one selected from a quinone derivative, a metal oxide, and a cyano group-containing compound, but embodiments of the present disclosure are not limited thereto.

(145) In one embodiment, the p-dopant may include at least one selected from: a quinone derivative, such as tetracyanoquinodimethane (TCNQ) or 2,3,5,6-tetrafluoro-7,7,8,8-tetracyanoquinodimethane (F4-TCNQ); a metal oxide, such as tungsten oxide or molybdenum oxide; 1,4,5,8,9,12-hexaazatriphenylene-hexacarbonitrile (HAT-CN); and a compound represented by Formula 221 below: but embodiments of the present disclosure are not limited thereto:

(146) ##STR00072##

(147) In Formula 221, R.sub.221 to R.sub.223 may each independently be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, wherein at least one selected from R.sub.221 to R.sub.223 may have at least one substituent selected from a cyano group, —F, —Cl, —Br, —I, a C.sub.1-C.sub.20 alkyl group substituted with —F, a C.sub.1-C.sub.20 alkyl group substituted with —Cl, a C.sub.1-C.sub.20 alkyl group substituted with —Br, and a C.sub.1-C.sub.20 alkyl group substituted with —

I.
Emission Layer in Organic Layer **150**

(148) When the organic light-emitting device **10** is a full-color organic light-emitting device, the emission layer may be patterned into a red emission layer, a green emission layer, or a blue emission layer, according to a sub-pixel. In one or more embodiments, the emission layer may have a stacked structure of two or more layers selected from a red emission layer, a green emission layer, and a blue emission layer, in which the two or more layers contact each other or are separated from each other. In one or more embodiments, the emission layer may include two or more materials selected from a red light-emitting material, a green light-emitting material, and a blue light-emitting material, in which the two or more materials are mixed with each other in a single layer to emit white light.

(149) The emission layer may include a host and a dopant. The dopant may include at least one selected from a phosphorescent dopant and a fluorescent dopant.

(150) In the emission layer, an amount of the dopant may be in a range of about 0.01 parts by weight to about 15 parts by weight based on 100 parts by weight of the host, but embodiments of the present disclosure are not limited thereto.

(151) A thickness of the emission layer may be in a range of about 100 Å to about 1,000 Å, for example, about 200 Å to about 600 Å. When the thickness of the emission layer is within this range, excellent light-emission characteristics may be obtained without a substantial increase in driving voltage.

(152) Host in Emission Layer

(153) In one or more embodiments, the host may include a compound represented by Formula 301 below.

[Ar.sub.301].sub.xb11—[(L.sub.301).sub.xb1-R.sub.301].sub.xb21 Formula 301

(154) In Formula 301, Ar.sub.301 may be a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, xb11 may be 0, 1, 2, or 3; L.sub.301 may be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group, xb1 may be an integer from 0 to 5, R.sub.301 may be selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.301)(Q.sub.302)(Q.sub.303), —N(Q.sub.301)(Q.sub.302), —B(Q.sub.301)(Q.sub.302), —C(=O)(Q.sub.301), —S(=O).sub.2(Q.sub.301), and —P(=O)(Q.sub.301)(Q.sub.302), xb21 may be an integer from 1 to 5, and Q.sub.301 to Q.sub.303 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group, but embodiments of the present disclosure are not limited

thereto.

(155) In one embodiment, Ar.sub.301 in Formula 301 may be selected from: a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, and a dibenzothiophene group; and a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, and a dibenzothiophene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.31 to Q.sub.33 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, an alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group, but embodiments of the present disclosure are not limited thereto.

(156) When xb11 in Formula 301 is 2 or more, two or more Ar.sub.301(s) may be linked to each other via a single bond.

(157) In one or more embodiments, the compound represented by Formula 301 may be represented by Formula 301-1 or 301-2 below:

(158) ##STR00073##

(159) In Formulae 301-1 and 301-2, A.sub.301 to A.sub.304 may each independently be selected from a benzene ring, a naphthalene ring, a phenanthrene ring, a fluoranthene ring, a triphenylene ring, a pyrene ring, a chrysene ring, a pyridine ring, a pyrimidine ring, an indene ring, a fluorene ring, a spiro-bifluorene ring, a benzofluorene ring, a dibenzofluorene ring, an indole ring, a carbazole ring, a benzocarbazole ring, a dibenzocarbazole ring, a furan ring, a benzofuran ring, a dibenzofuran ring, a naphthofuran ring, a benzonaphthofuran ring, a dinaphthofuran ring, a thiophene ring, a benzothiophene ring, a dibenzothiophene ring, a naphthothiophene ring, a benzonaphthothiophene ring, and a dinaphthothiophene ring, X.sub.301 may be O, S, or N—[(L.sub.304).sub.xb4-R.sub.304], R.sub.311 to R.sub.314 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), xb22 and xb23 may each independently be 0, 1, or 2, L.sub.301, xb1, R.sub.301, and Q.sub.31 to Q.sub.33 are the same as described herein above, L.sub.302 to L.sub.304 may each independently be the same as described in connection with L.sub.301, xb2 to xb4 may each independently be the same as described in connection with xb1, and R.sub.302 to R.sub.304 may each independently be the same as described in connection with R.sub.301.

(160) For example, L.sub.301 to L.sub.304 in Formulae 301, 301-1, and 301-2 may each independently be selected from:

(161) a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an

indolylene group, an isoindolylene group, a benzofuranylene group, a benzothiophenylene group, a dibenzofuranylene group, a dibenzothiophenylene group, a benzocarbazolylene group, a dibenzocarbazolylene group, a dibenzosilolylene group, a pyridinylene group, an imidazolylene group, a pyrazolylene group, a thiazolylene group, an isothiazolylene group, an oxazolylene group, an isoxazolylene group, a thiadiazolylene group, an oxadiazolylene group, a pyrazinylene group, a pyrimidinylene group, a pyridazinylene group, a triazinylene group, a quinolinylene group, an isoquinolinylene group, a benzoquinolinylene group, a phthalazinylene group, a naphthyridinylene group, a quinoxalinylene group, a quinazolinylene group, a cinnolinylene group, a phenanthridinylene group, an acridinylene group, a phenanthrolinylene group, a phenazinylene group, a benzimidazolylene group, an isobenzothiazolylene group, a benzoxazolylene group, an isobenzoxazolylene group, a triazolylene group, a tetrazolylene group, an imidazopyridinylene group, an imidazopyrimidinylene group, and an azacarbazolylene group; and a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenyene group, a fluoranthenyene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenyene group, a hexacenyene group, a pentacenyene group, a thiophenyene group, a furanylene group, a carbazolylene group, an indolylene group, an isoindolylene group, a benzofuranylene group, a benzothiophenylene group, a dibenzofuranylene group, a dibenzothiophenylene group, a benzocarbazolylene group, a dibenzocarbazolylene group, a dibenzosilolylene group, a pyridinylene group, an imidazolylene group, a pyrazolylene group, a thiazolylene group, an isothiazolylene group, an oxazolylene group, an isoxazolylene group, a thiadiazolylene group, an oxadiazolylene group, a pyrazinylene group, a pyrimidinylene group, a pyridazinylene group, a triazinylene group, a quinolinylene group, an isoquinolinylene group, a benzoquinolinylene group, a phthalazinylene group, a naphthyridinylene group, a quinoxalinylene group, a quinazolinylene group, a cinnolinylene group, a phenanthridinylene group, an acridinylene group, a phenanthrolinylene group, a phenazinylene group, a benzimidazolylene group, an isobenzothiazolylene group, a benzoxazolylene group, an isobenzoxazolylene group, a triazolylene group, a tetrazolylene group, an imidazopyridinylene group, an imidazopyrimidinylene group, and an azacarbazolylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.31 to Q.sub.33 are the same as described herein above.

(162) In one embodiment, R.sub.301 to R.sub.304 in Formulae 301, 301-1, and 301-2 may each

independently be selected from: a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group; and a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.31 to Q.sub.33 are the same as described herein above.

(163) In one or more embodiments, the host may include an alkaline earth metal complex. For example, the host may be selected from a Be complex (for example, Compound H55), a Mg complex, and a Zn complex.

(164) The host may include at least one selected from 9,10-di(2-naphthyl)anthracene (ADN), 2-methyl-9,10-bis(naphthalen-2-yl)anthracene (MADN), 9,10-di-(2-naphthyl)-2-t-butyl-anthracene (TBADN), 4,4'-bis(N-carbazolyl)-1,1'-biphenyl (CBP), 1,3-di-9-carbazolylbenzene (mCP), 1,3,5-tri(carbazol-9-yl)benzene (TCP), and Compounds H1 to H55, but embodiments of the present disclosure are not limited thereto:

(165) ##STR00074## ##STR00075## ##STR00076## ##STR00077## ##STR00078##
##STR00079## ##STR00080## ##STR00081## ##STR00082## ##STR00083## ##STR00084##
##STR00085##

Phosphorescent Dopant Included in Emission Layer in Organic Layer 150

(166) The dopant may include the organometallic compound represented by Formula 1.

(167) The phosphorescent dopant may include an organometallic complex represented by Formula 401 below:

(168) ##STR00086##

(169) In Formulae 401 and 402, M may be selected from iridium (Ir), platinum (Pt), palladium (Pd), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), rhodium (Rh), and thulium (Tm), L.sub.401 may be selected from ligands represented by Formula 402, and xc1 may be 1, 2, or 3, wherein, when xc1 is 2 or more, two or more L.sub.401(s) may be identical to or different from each other, L.sub.402 may be an organic ligand, and xc2 may be an integer from 0 to 4, wherein, when xc2 is 2 or more, two or more L.sub.402(s) may be identical to or different from each other, X.sub.401 to X.sub.404 may each independently be nitrogen or carbon, X.sub.401 and X.sub.403 may be linked via a single bond or a double bond, and X.sub.402 and X.sub.404 may be linked via a single bond or a double bond, A.sub.401 and A.sub.402 may each independently be selected from a C.sub.5-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, X.sub.405 may be a single bond, *—O—', *—S—*', *—C(=O)—*', *—N(Q.sub.411)—*', *—C(Q.sub.411)(Q.sub.412)—*', *—C(Q.sub.411)=C(Q.sub.412)—*', *—C(Q.sub.411)=*', or *=C(Q.sub.411)=*', wherein Q.sub.411 and Q.sub.412 may be hydrogen, deuterium, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group, X.sub.406 may be a single bond, O, or S, R.sub.401 and R.sub.402 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C.sub.1-C.sub.20 alkyl group, a substituted or unsubstituted C.sub.1-C.sub.20 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.401)(Q.sub.402)(Q.sub.403), —N(Q.sub.401)(Q.sub.402), —B(Q.sub.401)(Q.sub.402), —C(=O)(Q.sub.401), —S(=O).sub.2(Q.sub.401), and —P(=O)(Q.sub.401)(Q.sub.402), wherein Q.sub.401 to Q.sub.403 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a C.sub.6-C.sub.20 aryl group, and a C.sub.1-C.sub.20 heteroaryl group, xc11 and xc12 may each independently be an integer from 0 to 10, and '*' in Formula 402 each indicate a binding site to M in Formula 401.

(170) In one embodiment, A.sub.401 and A.sub.402 in Formula 402 may each independently be selected from a benzene group, a naphthalene group, a fluorene group, a spiro-bifluorene group, an

indene group, a pyrrole group, a thiophene group, a furan group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a quinoxaline group, a quinazoline group, a carbazole group, a benzimidazole group, a benzofuran group, a benzothiophene group, an isobenzothiophene group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a dibenzofuran group, and a dibenzothiophene group.

(171) In one or more embodiments, in Formula 402, i) X.sub.401 may be nitrogen, and X.sub.402 may be carbon, or ii) X.sub.401 and X.sub.402 may each be nitrogen at the same time.

(172) In one or more embodiments, R.sub.401 and R.sub.402 in Formula 501 may each independently be selected from: hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, and a C.sub.1-C.sub.20 alkoxy group; a C.sub.1-C.sub.20 alkyl group and a C.sub.1-C.sub.20 alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a phenyl group, a naphthyl group, a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, and a norbornenyl group; a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group; a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group; and —Si(Q.sub.401)(Q.sub.402)(Q.sub.403), —N(Q.sub.401)(Q.sub.402), —B(Q.sub.401)(Q.sub.402), —C(=O)(Q.sub.401), —S(=O).sub.2(Q.sub.401), and —P(=O)(Q.sub.401)(Q.sub.402), and Q.sub.401 to Q.sub.403 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, and a naphthyl group, but embodiments of the present disclosure are not limited thereto.

(173) In one or more embodiments, in Formula 401, when xc1 is 2 or more, two A.sub.401(s) in two or more L.sub.401(s) may optionally be linked via X.sub.407, which is a linking group, or two A.sub.402(s) in two or more L.sub.401(s) may optionally be linked via X.sub.408, which is a linking group (see Compounds PD1 to PD4 and PD7). X.sub.407 and X.sub.408 may each independently be a single bond, *—O—*, *—S—*, *—C(=O)—*, *—N(Q.sub.413)—*, *—C(Q.sub.413)(Q.sub.414)—*, or *—C(Q.sub.413)=C(Q.sub.414)—* (wherein Q.sub.413 and Q.sub.414 may each independently be hydrogen, deuterium, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group), but embodiments of the present disclosure are not limited thereto.

(174) L.sub.402 in Formula 401 may be a monovalent, divalent, or trivalent organic ligand. For example, L.sub.402 may be selected from halogen, diketone (for example, acetylacetonate),

carboxylic acid (for example, picolinate), —C(=O) , isonitrile, —CN , and phosphorus (for example, phosphine, or phosphite), but embodiments of the present disclosure are not limited thereto.

(175) In one or more embodiments, the phosphorescent dopant may be selected from, for example, Compounds PD1 to PD25, but embodiments of the present disclosure are not limited thereto:

(176) ##STR00087## ##STR00088## ##STR00089## ##STR00090## ##STR00091##
##STR00092##

Fluorescent Dopant in Emission Layer

(177) The fluorescent dopant may include an arylamine compound or a styrylamine compound.

(178) The fluorescent dopant may include a compound represented by Formula 501 below:

(179) ##STR00093##

(180) In Formula 501, Ar.sub.501 may be a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, L.sub.501 to L.sub.503 may each independently be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group, xd1 to xd3 may each independently be an integer from 0 to 3, R.sub.501 and R.sub.502 may each independently be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, and xd4 may be an integer from 1 to 6.

(181) In one embodiment, Ar.sub.501 in Formula 501 may be selected from: a naphthalene group, a heptalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, and an indenophenanthrene group; and a naphthalene group, a heptalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, and an indenophenanthrene group, each substituted with at least one selected from deuterium, —F , —Cl , —Br , —I , a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

(182) In one or more embodiments, L.sub.501 to L.sub.503 in Formula 501 may each independently be selected from: a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylene group, a pentacenylene group, a thiophenylenylene group, a furanylene group, a

carbazolylene group, an indolylene group, an isoindolylene group, a benzofuranylene group, a benzothiophenylylene group, a dibenzofuranylene group, a dibenzothiophenylylene group, a benzocarbazolylylene group, a dibenzocarbazolylylene group, a dibenzosilolylylene group, and a pyridinylylene group; and a phenylene group, a naphthylylene group, a fluorenylylene group, a spiro-bifluorenylylene group, a benzofluorenylylene group, a dibenzofluorenylylene group, a phenanthrenylylene group, an anthracenylylene group, a fluoranthenylylene group, a triphenylenylylene group, a pyrenylylene group, a chrysenylylene group, a perylenylylene group, a pentaphenylylene group, a hexacenylylene group, a pentacenylylene group, a thiophenylylene group, a furanylylene group, a carbazolylylene group, an indolylylene group, an isoindolylylene group, a benzofuranylylene group, a benzothiophenylylene group, a dibenzofuranylylene group, a dibenzothiophenylylene group, a benzocarbazolylylene group, a dibenzocarbazolylylene group, a dibenzosilolylylene group, and a pyridinylylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group.

(183) In one or more embodiments, R.sub.501 and R.sub.502 in Formula 501 may each independently be selected from: a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group; and a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group and —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), and Q.sub.31 to Q.sub.33 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

(184) In one or more embodiments, xd4 in Formula 501 may be 2, but embodiments of the present

disclosure are not limited thereto.

(185) For example, the fluorescent dopant may be selected from Compounds FD1 to FD22:

(186) ##STR00094## ##STR00095## ##STR00096## ##STR00097## ##STR00098##
##STR00099##

(187) In one or more embodiments, the fluorescent dopant may be selected from the following compounds, but embodiments of the present disclosure are not limited thereto:

(188) ##STR00100##

Electron Transport Region in Organic Layer 150

(189) The electron transport region may have i) a single-layered structure including a single layer including a single material, ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

(190) The electron transport region may include at least one selected from a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, and an electron injection layer, but embodiments of the present disclosure are not limited thereto.

(191) For example, the electron transport region may have an electron transport layer/electron injection layer structure, a hole blocking layer/electron transport layer/electron injection layer structure, an electron control layer/electron transport layer/electron injection layer structure, or a buffer layer/electron transport layer/electron injection layer structure, wherein for each structure, constituting layers are sequentially stacked from an emission layer. However, embodiments of the structure of the electron transport region are not limited thereto.

(192) The electron transport region (for example, a buffer layer, a hole blocking layer, an electron control layer, or an electron transport layer in the electron transport region) may include a metal-free compound containing at least one π electron-depleted nitrogen-containing ring.

(193) The “ π electron-depleted nitrogen-containing ring” indicates a C.sub.1-C.sub.60 heterocyclic group having at least one —N= moiety as a ring-forming moiety.

(194) For example, the “ π electron-depleted nitrogen-containing ring” may be i) a 5-membered to π -membered heteromonocyclic group having at least one —N= moiety, ii) a heteropolycyclic group in which two or more 5-membered to π -membered heteromonocyclic groups each having at least one —N= moiety are condensed with each other, or iii) a heteropolycyclic group in which at least one of 5-membered to π -membered heteromonocyclic groups, each having at least one —N= moiety, is condensed with at least one C.sub.5-C.sub.60 carbocyclic group.

(195) Examples of the π electron-depleted nitrogen-containing ring include an imidazole, a pyrazole, a thiazole, an isothiazole, an oxazole, an isoxazole, a pyridine, a pyrazine, a pyrimidine, a pyridazine, an indazole, a purine, a quinoline, an isoquinoline, a benzoquinoline, a phthalazine, a naphthyridine, a quinoxaline, a quinazoline, a cinnoline, a phenanthridine, an acridine, a phenanthroline, a phenazine, a benzimidazole, an isobenzothiazole, a benzoxazole, an isobenzoxazole, a triazole, a tetrazole, an oxadiazole, a triazine, a thiadiazole, an imidazopyridine, an imidazopyrimidine, and an azacarbazole, but are not limited thereto.

(196) For example, the electron transport region may include a compound represented by Formula 601:

[Ar.sub.601].sub.xe11-[(L.sub.601).sub.xe1-R.sub.601]).sub.xe21 Formula 601

(197) In Formula 601, Ar.sub.601 may be a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, xe11 may be 0, 1, 2, or 3, L.sub.601 may be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkylene group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenylene group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenylene group, a substituted or unsubstituted C.sub.6-C.sub.60 arylene group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted

divalent non-aromatic condensed heteropolycyclic group, xe1 may be an integer from 0 to 5, R.sub.601 may be selected from a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.601)(Q.sub.602)(Q.sub.603), —C(=O)(Q.sub.601), —S(=O).sub.2(Q.sub.601), and —P(=O)(Q.sub.601)(Q.sub.602), Q.sub.601 to Q.sub.603 may each independently be a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group, and xe21 may be an integer from 1 to 5.

(198) In one embodiment, at least one of Ar.sub.601(s) in the number of xe11 and R.sub.601(s) in the number of xe21 may include the π electron-depleted nitrogen-containing ring.

(199) In one embodiment, Ar.sub.601 in Formula 601 may be selected from: a benzene group, a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, a dibenzothiophene group, a carbazole group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, an indazole group, a purine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a phthalazine group, a naphthyridine group, a quinoxaline group, a quinazoline group, a cinnoline group, a phenanthridine group, an acridine group, a phenanthroline group, a phenazine group, a benzimidazole group, an isobenzothiazole group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a thiadiazole group, an imidazopyridine group, an imidazopyrimidine group, and an azacarbazole group; and a benzene group, a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, a dibenzothiophene group, a carbazole group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, an indazole group, a purine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a phthalazine group, a naphthyridine group, a quinoxaline group, a quinazoline group, a cinnoline group, a phenanthridine group, an acridine group, a phenanthroline group, a phenazine group, a benzimidazole group, an isobenzothiazole group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a thiadiazole group, an imidazopyridine group, an imidazopyrimidine group, and an azacarbazole group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.31 to Q.sub.33 may each independently be selected from a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

(200) When xe11 in Formula 601 is 2 or more, two or more Ar.sub.601(s) may be linked to each other via a single bond.

(201) In one or more embodiments, Ar.sub.601 in Formula 601 may be an anthracene group.

(202) In one or more embodiments, a compound represented by Formula 601 may be represented by Formula 601-1 below:

(203) ##STR00101##

(204) In Formula 601-1, X.sub.614 may be N or C(R.sub.614), X.sub.615 may be N or C(R.sub.615), and X.sub.616 may be N or C(R.sub.616), wherein at least one selected from X.sub.614 to X.sub.616 may be N, L.sub.611 to L.sub.613 may each independently be the same as described in connection with L.sub.601, xe611 to xe613 may each independently be defined the same as xe1, R.sub.611 to R.sub.613 may each independently be the same as described in connection with R.sub.601, and R.sub.614 to R.sub.616 may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

(205) In one embodiment, L.sub.601 and L.sub.611 to L.sub.613 in Formulae 601 and 601-1 may each independently be selected from:

(206) a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group; and a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a

hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group, but embodiments of the present disclosure are not limited thereto.

(207) In one or more embodiments, xe1 and xe611 to xe613 in Formulae 601 and 601-1 may each independently be 0, 1, or 2.

(208) In one or more embodiments, R.sub.601 and R.sub.611 to R.sub.613 in Formulae 601 and 601-1 may each independently be selected from: a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group; a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group, each

substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group; and —S(=O).sub.2(Q.sub.601) and —P(=O)(Q.sub.601) (Q.sub.602), and Q.sub.601 and Q.sub.602 are the same as described herein above.

(209) The electron transport region may include at least one compound selected from Compounds ET1 to ET36, but embodiments of the present disclosure are not limited thereto:

(210) ##STR00102## ##STR00103## ##STR00104## ##STR00105## ##STR00106##
##STR00107## ##STR00108## ##STR00109## ##STR00110## ##STR00111## ##STR00112##
##STR00113##

(211) In one or more embodiments, the electron transport region may include at least one compound selected from 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), Alq.sub.3, BAlq, 3-(biphenyl-4-yl)-5-(4-tert-butylphenyl)-4-phenyl-4H-1,2,4-triazole (TAZ), and NTAZ:

(212) ##STR00114##

(213) Thicknesses of the buffer layer, the hole blocking layer, and the electron control layer may each be in a range of about 20 Å to about 1,000 Å, for example, about 30 Å to about 300 Å. When the thicknesses of the buffer layer, the hole blocking layer, and the electron control layer are within these ranges, the buffer layer, the hole blocking layer, and the electron control layer may have excellent hole blocking characteristics or electron control characteristics without a substantial increase in driving voltage.

(214) A thickness of the electron transport layer may be in a range of about 100 Å to about 1,000 Å, for example, about 150 Å to about 500 Å. When the thickness of the electron transport layer is within the range described herein above, the electron transport layer may have suitable or satisfactory electron transport characteristics without a substantial increase in driving voltage.

(215) The electron transport region (for example, the electron transport layer in the electron transport region) may further include, in addition to the materials described herein above, a metal-containing material.

(216) The metal-containing material may include at least one selected from alkali metal complex and alkaline earth-metal complex. The alkali metal complex may include a metal ion selected from a Li ion, a Na ion, a K ion, a Rb ion, and a Cs ion, and the alkaline earth-metal complex may include a metal ion selected from a Be ion, a Mg ion, a Ca ion, a Sr ion, and a Ba ion. A ligand coordinated with the metal ion of the alkali metal complex or the alkaline earth-metal complex may be selected from a hydroxy quinoline, a hydroxy isoquinoline, a hydroxy benzoquinoline, a hydroxy acridine, a hydroxy phenanthridine, a hydroxy phenyloxazole, a hydroxy phenylthiazole, a hydroxy diphenyloxadiazole, a hydroxy diphenylthiadiazole, a hydroxy phenylpyridine, a hydroxy

phenylbenzimidazole, a hydroxy phenylbenzothiazole, a bipyridine, a phenanthroline, and a cyclopentadiene, but embodiments of the present disclosure are not limited thereto.

(217) For example, the metal-containing material may include a Li complex. The Li complex may include, for example, Compound ET-D1 (lithium quinolate, LiQ) or ET-D2.

(218) ##STR00115##

(219) The electron transport region may include an electron injection layer that facilitates electron injection from the second electrode **190**. The electron injection layer may directly contact the second electrode **190**.

(220) The electron injection layer may have i) a single-layered structure including a single layer including a single material, ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

(221) The electron injection layer may include an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combinations thereof.

(222) The alkali metal may be selected from Li, Na, K, Rb, and Cs. In one embodiment, the alkali metal may be Li, Na, or Cs. In one or more embodiments, the alkali metal may be Li or Cs, but embodiments of the present disclosure are not limited thereto.

(223) The alkaline earth metal may be selected from Mg, Ca, Sr, and Ba.

(224) The rare earth metal may be selected from Sc, Y, Ce, Tb, Yb, and Gd.

(225) The alkali metal compound, the alkaline earth-metal compound, and the rare earth metal compound may be selected from oxides and halides (for example, fluorides, chlorides, bromides, or iodides) of the alkali metal, the alkaline earth-metal, and the rare earth metal.

(226) The alkali metal compound may be selected from alkali metal oxides, such as Li_2O , Cs_2O , or K_2O , and alkali metal halides, such as LiF, NaF, CsF, KF, LiI, NaI, CsI, or KI. In one embodiment, the alkali metal compound may be selected from LiF, Li_2O , NaF, LiI, NaI, CsI, and KI, but embodiments of the present disclosure are not limited thereto.

(227) The alkaline earth-metal compound may be selected from alkaline earth-metal oxides, such as BaO, SrO, CaO, $\text{Ba}_x\text{Sr}_{1-x}\text{O}$ ($0 < x < 1$), or $\text{Ba}_x\text{Ca}_{1-x}\text{O}$ ($0 < x < 1$). In one embodiment, the alkaline earth-metal compound may be selected from BaO, SrO, and CaO, but embodiments of the present disclosure are not limited thereto.

(228) The rare earth metal compound may be selected from YbF_3 , ScF_3 , Sc_2O_3 , Y_2O_3 , Ce_2O_3 , GdF_3 , and TbF_3 . In one embodiment, the rare earth metal compound may be selected from YbF_3 , ScF_3 , TbF_3 , YbI_3 , ScI_3 , and TbI_3 , but embodiments of the present disclosure are not limited thereto.

(229) The alkali metal complex, the alkaline earth-metal complex, and the rare earth metal complex may include an ion of alkali metal, alkaline earth-metal, and rare earth metal as described herein above, and a ligand coordinated with a metal ion of the alkali metal complex, the alkaline earth-metal complex, or the rare earth metal complex may be selected from hydroxy quinoline, hydroxy isoquinoline, hydroxy benzoquinoline, hydroxy acridine, hydroxy phenanthridine, hydroxy phenyloxazole, hydroxy phenylthiazole, hydroxy diphenyloxadiazole, hydroxy diphenylthiadiazole, hydroxy phenylpyridine, hydroxy phenylbenzimidazole, hydroxy phenylbenzothiazole, bipyridine, phenanthroline, and cyclopentadiene, but embodiments of the present disclosure are not limited thereto.

(230) The electron injection layer may consist of an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combinations thereof, as described herein above. In one or more embodiments, the electron injection layer may further include an organic material. When the electron injection layer further

includes an organic material, an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combinations thereof may be homogeneously or non-homogeneously dispersed in a matrix including the organic material.

(231) A thickness of the electron injection layer may be in a range of about 1 Å to about 100 Å, for example, about 3 Å to about 90 Å. When the thickness of the electron injection layer is within the range described herein above, the electron injection layer may have suitable or satisfactory electron injection characteristics without a substantial increase in driving voltage.

(232) Second Electrode **190**

(233) The second electrode **190** may be on the organic layer **150** having such a structure. The second electrode **190** may be a cathode which is an electron injection electrode, and in this regard, a material for forming the second electrode **190** may be selected from metal, an alloy, an electrically conductive compound, and a combination thereof, which have a relatively low work function.

(234) The second electrode **190** may include at least one selected from lithium (Li), silver (Ag), magnesium (Mg), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), ITO, and IZO, but embodiments of the present disclosure are not limited thereto. The second electrode **190** may be a transmissive electrode, a semi-transmissive electrode, or a reflective electrode.

(235) The second electrode **190** may have a single-layered structure, or a multi-layered structure including two or more layers.

(236) Description of FIGS. **2** to **4**

(237) FIG. **2** is a schematic view of an organic light-emitting device **20** according to an embodiment. The organic light-emitting device **20** includes a first capping layer **210**, the first electrode **110**, the organic layer **150**, and the second electrode **190**, which are sequentially stacked in this stated order. FIG. **3** is a schematic view of an organic light-emitting device **30** according to an embodiment. The organic light-emitting device **30** includes the first electrode **110**, the organic layer **150**, the second electrode **190**, and a second capping layer **220**, which are sequentially stacked in this stated order. FIG. **4** is a schematic view of an organic light-emitting device **40** according to an embodiment. The organic light-emitting device **40** includes the first capping layer **210**, the first electrode **110**, the organic layer **150**, the second electrode **190**, and the second capping layer **220**, which are sequentially stacked in this stated order.

(238) Regarding FIGS. **2** to **4**, the first electrode **110**, the organic layer **150**, and the second electrode **190** may be understood by referring to the description presented in connection with FIG. **1**.

(239) In the organic layer **150** of each of the organic light-emitting devices **20** and **40**, light generated in an emission layer may pass through the first electrode **110** and the first capping layer **210** toward the outside, wherein the first electrode **110** may be a semi-transmissive electrode or a transmissive electrode. In the organic layer **150** of each of the organic light-emitting devices **30** and **40**, light generated in an emission layer may pass through the second electrode **190** and the second capping layer **220** toward the outside, wherein the second electrode **190** may be a semi-transmissive electrode or a transmissive electrode.

(240) The first capping layer **210** and the second capping layer **220** may increase external luminescent efficiency according to the principle of constructive interference.

(241) The first capping layer **210** and the second capping layer **220** may each independently be an organic capping layer including an organic material, an inorganic capping layer including an inorganic material, or a composite capping layer including an organic material and an inorganic material.

(242) At least one selected from the first capping layer **210** and the second capping layer **220** may

each independently include at least one material selected from carbocyclic compounds, heterocyclic compounds, amine-based compounds, porphyrine derivatives, phthalocyanine derivatives, a naphthalocyanine derivatives, alkali metal complexes, and alkaline earth-based complexes. The carbocyclic compound, the heterocyclic compound, and the amine-based compound may be optionally substituted with a substituent containing at least one element selected from O, N, S, Se, Si, F, Cl, Br, and I. In one embodiment, at least one selected from the first capping layer **210** and the second capping layer **220** may each independently include an amine-based compound.

(243) In one embodiment, at least one selected from the first capping layer **210** and the second capping layer **220** may each independently include the compound represented by Formula 201 or the compound represented by Formula 202.

(244) In one or more embodiments, at least one selected from the first capping layer **210** and the second capping layer **220** may each independently include a compound selected from Compounds HT28 to HT33 and Compounds CP1 to CP5, but embodiments of the present disclosure are not limited thereto:

(245) ##STR00116##

(246) Hereinbefore, the organic light-emitting device according to an embodiment has been described in connection with FIGS. **1** to **4**, but embodiments of the present disclosure are not limited thereto.

(247) Layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region may be formed in a set or certain region by using one or more suitable methods selected from vacuum deposition, spin coating, casting, Langmuir-Blodgett (LB) deposition, ink-jet printing, laser-printing, and laser-induced thermal imaging.

(248) When layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region are formed by vacuum deposition, the deposition may be performed at a deposition temperature of about 100° C. to about 500° C., a vacuum degree of about 10.^{sup.}–8 torr to about 10.^{sup.}–3 torr, and a deposition speed of about 0.01 Å/sec to about 100 Å/sec by taking into account a material to be included in a layer to be formed, and the structure of a layer to be formed.

(249) When layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region are formed by spin coating, the spin coating may be performed at a coating speed of about 2,000 rpm to about 5,000 rpm and at a heat treatment temperature of about 80° C. to about 200° C. by taking into account a material to be included in a layer to be formed, and the structure of a layer to be formed.

(250) General Definition of at Least Some of the Substituents

(251) The term “C.sub.1-C.sub.60 alkyl group,” as used herein, refers to a linear or branched aliphatic saturated hydrocarbon monovalent group having 1 to 60 carbon atoms, and examples thereof include a methyl group, an ethyl group, a propyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, a pentyl group, an isoamyl group, and a hexyl group. The term “C.sub.1-C.sub.60 alkylene group,” as used herein, refers to a divalent group having substantially the same structure as the C.sub.1-C.sub.60 alkyl group.

(252) The term “C.sub.2-C.sub.60 alkenyl group,” as used herein, refers to a hydrocarbon group having at least one carbon-carbon double bond in the middle or at the terminus of the C.sub.2-C.sub.60 alkyl group, and examples thereof include an ethenyl group, a propenyl group, and a butenyl group. The term “C.sub.2-C.sub.60 alkenylene group,” as used herein, refers to a divalent group having substantially the same structure as the C.sub.2-C.sub.60 alkenyl group.

(253) The term “C.sub.2-C.sub.60 alkynyl group,” as used herein, refers to a hydrocarbon group having at least one carbon-carbon triple bond at a main chain (e.g., in the middle) or at a terminal end (e.g., the terminus) of the C.sub.2-C.sub.60 alkyl group, and examples thereof include an ethynyl group, and a propynyl group. The term “C.sub.2-C.sub.60 alkynylene group,” as used

herein, refers to a divalent group having substantially the same structure as the C.sub.2-C.sub.60 alkynyl group.

(254) The term “C.sub.1-C.sub.60 alkoxy group,” as used herein, refers to a monovalent group represented by —OA.sub.101 (wherein A.sub.101 is the C.sub.1-C.sub.60 alkyl group), and examples thereof include a methoxy group, an ethoxy group, and an isopropoxy group.

(255) The term “C.sub.3-C.sub.10 cycloalkyl group,” as used herein, refers to a monovalent saturated hydrocarbon monocyclic group having 3 to 10 carbon atoms, and examples thereof include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, and a cycloheptyl group. The term “C.sub.3-C.sub.10 cycloalkylene group,” as used herein, refers to a divalent group having substantially the same structure as the C.sub.3-C.sub.10 cycloalkyl group.

(256) The term “C.sub.1-C.sub.10 heterocycloalkyl group,” as used herein, refers to a monovalent monocyclic group having at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom and 1 to 10 carbon atoms, and examples thereof include a 1,2,3,4-oxatriazolidinyl group, a tetrahydrofuranyl group, and a tetrahydrothiophenyl group. The term “C.sub.1-C.sub.10 heterocycloalkylene group,” as used herein, refers to a divalent group having substantially the same structure as the C.sub.1-C.sub.10 heterocycloalkyl group.

(257) The term “C.sub.3-C.sub.10 cycloalkenyl group,” as used herein, refers to a monovalent monocyclic group that has 3 to 10 carbon atoms and at least one carbon-carbon double bond in the ring thereof and no aromaticity (e.g., is not aromatic), and examples thereof include a cyclopentenyl group, a cyclohexenyl group, and a cycloheptenyl group. The term “C.sub.3-C.sub.10 cycloalkenylene group,” as used herein, refers to a divalent group having substantially the same structure as the C.sub.3-C.sub.10 cycloalkenyl group.

(258) The term “C.sub.1-C.sub.10 heterocycloalkenyl group,” as used herein, refers to a monovalent monocyclic group that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, 1 to 10 carbon atoms, and at least one carbon-carbon double bond in its ring. Non-limiting examples of the C.sub.1-C.sub.10 heterocycloalkenyl group include a 4,5-dihydro-1,2,3,4-oxatriazolyl group, a 2,3-dihydrofuranyl group, and a 2,3-dihydrothiophenyl group. The term “C.sub.1-C.sub.10 heterocycloalkenylene group,” as used herein, refers to a divalent group having substantially the same structure as the C.sub.1-C.sub.10 heterocycloalkenyl group.

(259) The term “C.sub.6-C.sub.60 aryl group,” as used herein, refers to a monovalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms, and the term “C.sub.6-C.sub.60 arylene group,” as used herein, refers to a divalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms. Non-limiting examples of the C.sub.6-C.sub.60 aryl group include a phenyl group, a naphthyl group, an anthracenyl group, a phenanthrenyl group, a pyrenyl group, and a chrysenyl group. When the C.sub.6-C.sub.60 aryl group and the C.sub.6-C.sub.60 arylene group each include two or more rings, the rings may be fused to each other (e.g., combined together).

(260) The term “C.sub.1-C.sub.60 heteroaryl group,” as used herein, refers to a monovalent group having a carbocyclic aromatic system that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, in addition to 1 to 60 carbon atoms. The term “C.sub.1-C.sub.60 heteroarylene group,” as used herein, refers to a divalent group having a carbocyclic aromatic system that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, in addition to 1 to 60 carbon atoms. Non-limiting examples of the C.sub.1-C.sub.60 heteroaryl group include a pyridinyl group, a pyrimidinyl group, a pyrazinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, and an isoquinolinyl group. When the C.sub.1-C.sub.60 heteroaryl group and the C.sub.1-C.sub.60 heteroarylene group each include two or more rings, the rings may be condensed with each other (e.g., combined together).

(261) The term “C.sub.6-C.sub.60 aryloxy group,” as used herein, refers to —OA.sub.102 (wherein A.sub.102 is the C.sub.6-C.sub.60 aryl group), and the term “C.sub.6-C.sub.60 arylthio group,” as used herein, indicates —SA.sub.103 (wherein A.sub.103 is the C.sub.6-C.sub.60 aryl group).

(262) The term “monovalent non-aromatic condensed polycyclic group,” as used herein, refers to a

monovalent group (for example, having 8 to 60 carbon atoms) having two or more rings condensed with each other (e.g., combined together), only carbon atoms as ring-forming atoms, and containing at least one non-aromatic ring in its entire molecular structure. An example of the monovalent non-aromatic condensed polycyclic group is a fluorenyl group. The term “divalent non-aromatic condensed polycyclic group,” as used herein, refers to a divalent group having substantially the same structure as the monovalent non-aromatic condensed polycyclic group.

(263) The term “monovalent non-aromatic condensed heteropolycyclic group,” as used herein, refers to a monovalent group (for example, having 1 to 60 carbon atoms) having two or more rings condensed to each other (e.g., combined together), at least one heteroatom selected from N, O, Si, P, and S, other than carbon atoms, as a ring-forming atom, and containing at least one non-aromatic ring in its entire molecular structure. An example of the monovalent non-aromatic condensed heteropolycyclic group is a carbazolyl group. The term “divalent non-aromatic condensed heteropolycyclic group,” as used herein, refers to a divalent group having substantially the same structure as the monovalent non-aromatic condensed heteropolycyclic group.

(264) The term “C.sub.5-C.sub.60 carbocyclic group,” as used herein, refers to a monocyclic or polycyclic group having 5 to 60 carbon atoms in which a ring-forming atom is a carbon atom only. The term “C.sub.5-C.sub.60 carbocyclic group,” as used herein, refers to an aromatic carbocyclic group or a non-aromatic carbocyclic group. The C.sub.5-C.sub.60 carbocyclic group may be a ring, such as benzene, a monovalent group, such as a phenyl group, or a divalent group, such as a phenylene group. In one or more embodiments, depending on the number of substituents connected to the C.sub.5-C.sub.60 carbocyclic group, the C.sub.5-C.sub.60 carbocyclic group may be a trivalent group or a quadrivalent group.

(265) The term “C.sub.1-C.sub.60 heterocyclic group,” as used herein, refers to a group having substantially the same structure as the C.sub.5-C.sub.60 carbocyclic group, except that as a ring-forming atom, at least one heteroatom selected from N, O, Si, P, and S is used in addition to carbon (the number of carbon atoms may be in a range of 1 to 60).

(266) In the present specification, at least one substituent of the substituted C.sub.1-C.sub.60 heterocyclic group, the substituted C.sub.3-C.sub.10 cycloalkylene group, the substituted C.sub.1-C.sub.10 heterocycloalkylene group, the substituted C.sub.3-C.sub.10 cycloalkenylene group, the substituted C.sub.1-C.sub.10 heterocycloalkenylene group, the substituted C.sub.6-C.sub.60 arylene group, the substituted C.sub.1-C.sub.60 heteroarylene group, the substituted divalent non-aromatic condensed polycyclic group, the substituted divalent non-aromatic condensed heteropolycyclic group, the substituted C.sub.1-C.sub.60 alkyl group, the substituted C.sub.2-C.sub.60 alkenyl group, the substituted C.sub.2-C.sub.60 alkynyl group, the substituted C.sub.1-C.sub.60 alkoxy group, the substituted C.sub.3-C.sub.10 cycloalkyl group, the substituted C.sub.1-C.sub.10 heterocycloalkyl group, the substituted C.sub.3-C.sub.10 cycloalkenyl group, the substituted C.sub.1-C.sub.10 heterocycloalkenyl group, the substituted C.sub.6-C.sub.60 aryl group, the substituted C.sub.6-C.sub.60 aryloxy group, the substituted C.sub.6-C.sub.60 arylthio group, the substituted C.sub.1-C.sub.60 heteroaryl group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group may be selected from: deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, and a C.sub.1-C.sub.60 alkoxy group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, and a C.sub.1-C.sub.60 alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a

monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, $\text{—Si}(\text{C.sub.211})(\text{C.sub.212})(\text{Q.sub.13})$, $\text{—N}(\text{Q.sub.11})(\text{Q.sub.12})$, $\text{—B}(\text{Q.sub.11})(\text{Q.sub.12})$, $\text{—C(=O)}(\text{Q.sub.11})$, $\text{—S(=O).sub.2}(\text{Q.sub.11})$, and $\text{—P(=O)}(\text{Q.sub.11})(\text{Q.sub.12})$; a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.2-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.2-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.2-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group; a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group, each substituted with at least one selected from deuterium, —F , —Cl , —Br , —I , a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, $\text{—Si}(\text{Q.sub.21})(\text{Q.sub.22})(\text{Q.sub.23})$, $\text{—N}(\text{Q.sub.21})(\text{Q.sub.22})$, $\text{—B}(\text{Q.sub.21})(\text{Q.sub.22})$, $\text{—C(=O)}(\text{Q.sub.21})$, $\text{—S(=O).sub.2}(\text{Q.sub.21})$, and $\text{—P(=O)}(\text{Q.sub.21})(\text{Q.sub.22})$; and $\text{—Si}(\text{Q.sub.31})(\text{Q.sub.32})(\text{Q.sub.33})$, $\text{—N}(\text{Q.sub.31})(\text{Q.sub.32})$, $\text{—B}(\text{Q.sub.31})(\text{Q.sub.32})$, $\text{—C(=O)}(\text{Q.sub.31})$, $\text{—S(=O).sub.2}(\text{Q.sub.31})$, and $\text{—P(=O)}(\text{Q.sub.31})(\text{Q.sub.32})$, and Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 may each independently be selected from hydrogen, deuterium, —F , —Cl , —Br , —I , a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a biphenyl group, and a terphenyl group.

(267) The term “Ph,” as used herein, represents a phenyl group, the term “Me,” as used herein, represents a methyl group, the term “Et,” as used herein, represents an ethyl group, the term “ter-Bu” or “But,” as used herein, represents a tert-butyl group, and the term “OMe,” as used herein, represents a methoxy group.

(268) The term “biphenyl group,” as used herein, refers to a “phenyl group substituted with a phenyl group. The “biphenyl group” is a “substituted phenyl group” having a “C.sub.6-C.sub.60 aryl group” as a substituent.

(269) The term “terphenyl group,” as used herein, refers to a “phenyl group substituted with a biphenyl group. The “terphenyl group” is a “phenyl group” having, as a substituent, a “C.sub.6-C.sub.60 aryl group substituted with a C.sub.6-C.sub.60 aryl group”.

(270) * and *”, as used herein, unless defined otherwise, each refer to a binding site to a neighboring atom in a corresponding formula.

(271) Hereinafter, a compound according to embodiments and an organic light-emitting device according to embodiments will be described in more detail with reference to Synthesis Examples and Examples. The expression “B was used instead of A” used in describing Synthesis Examples means that an identical number of molar equivalents of B was used in place of molar equivalents of A.

SYNTHESIS EXAMPLES

Synthesis Example 1: Synthesis of Compound 1

(272) ##STR00117## ##STR00118##

(1) Synthesis of Intermediate [1-A]

(273) 3,5-dibromobenzonitrile (1.0 eq), imidazole (3.0 eq), CuI (0.02 eq), K.sub.2CO.sub.3 (3.0 eq), and L-Proline (0.04 eq) were dissolved in dimethylsulfonate (0.1 M) and stirred at a temperature of 160° C. for 48 hours. The reaction mixture was cooled to room temperature, and an organic layer was extracted therefrom three times using dichloromethane and water. The extracted organic layer was dried using magnesium sulfate and concentrated, and purified by column chromatography to synthesize Intermediate [1-A] (yield: 42%).

(274) (2) Synthesis of Intermediate [1-B]

(275) Intermediate [1-A] (1.0 eq) and diiodomethane (3.0 eq) were dissolved in toluene (1.0 M) and stirred at a temperature of 120° C. for 12 hours. The reaction mixture was cooled to room temperature, washed with toluene, and filtered to obtain a solid. The obtained solid was dried to synthesize Intermediate [1-B] (yield: 68%).

(276) (3) Synthesis of Intermediate [1-C]

(277) 2,6-dibromopyridine (1.0 eq), 9H-carbazole-3,6-dicarbonitrile (1.1 eq), CuI (0.01 eq), K.sub.2CO.sub.3 (2.0 eq), and L-Proline (0.02 eq) were dissolved in dimethylsulfonate (0.1 M) and stirred at a temperature of 160° C. for 48 hours. The reaction mixture was cooled to room temperature, and an organic layer was extracted therefrom three times using dichloromethane and water. The extracted organic layer was dried using magnesium sulfate and concentrated, and purified by column chromatography to synthesize Intermediate [1-C] (yield: 57%).

(278) (4) Synthesis of Intermediate [1-D]

(279) A reaction solution including 4,4,4-trifluoroacetamide methyl acetate (1.0 eq) and acetic acid (1.1 eq) was stirred at a temperature of 6° C., and hydrazine monohydrate (1.1 eq) was slowly added thereto for 0.3 hours. The reaction mixture was slowly heated to a temperature of 80° C. for 1 hour and stirred for 3 hours. After the reaction was believed to be completed, the reaction product was cooled to obtain Intermediate [1-D] (yield: 82%) in a crystal form.

(280) (5) Synthesis of Intermediate [1-E]

(281) Intermediate [1-C] (1.0 eq), Intermediate [1-D] (1.2 eq), CuI (0.01 eq), K.sub.2CO.sub.3 (2.0 eq), and L-Proline (0.02 eq) were dissolved in dimethylsulfonate (0.1 M) and stirred at a temperature of 160° C. for 48 hours. The reaction mixture was cooled to room temperature, and an organic layer was extracted therefrom three times using dichloromethane and water. The extracted organic layer was dried using magnesium sulfate and concentrated, and purified by column chromatography to synthesize Intermediate [1-E] (yield: 47%).

(282) (6) Synthesis of Compound 1

(283) Intermediate [1-B] (1.0 eq), Intermediate [1-E] (1.0 eq), iridium chloride hydrate (1.1 eq), and K.sub.2CO.sub.3 (10 eq) were dissolved in propionic acid (0.2 M) and stirred at a temperature of 140° C. for 12 hours. The reaction mixture was cooled to room temperature, and a solvent was removed therefrom under reduced pressure. Then, an organic layer was extracted therefrom three times using dichloromethane and water. The extracted organic layer was dried using magnesium sulfate and concentrated, and purified by column chromatography to synthesize Compound 1 (yield: 22%).

Synthesis Example 2: Synthesis of Compound 13

(284) ##STR00119## ##STR00120##

(1) Synthesis of Intermediate [13-A]

(285) Intermediate [13-A] (yield: 52%) was synthesized in substantially the same manner as in Synthesis of Intermediate [1-A], except that 3,5-dibromotrifluoromethane was used instead of 3,5-dibromobenzonitrile.

(286) (2) Synthesis of Intermediate [13-B]

(287) Intermediate [13-B] (yield: 80%) was synthesized in substantially the same manner as in

Synthesis of Intermediate [1-B], except that Intermediate [13-A] was used instead of Intermediate [1-A].

(288) (3) Synthesis of Intermediate [13-C]

(289) Intermediate [13-C] (yield: 67%) was synthesized in substantially the same manner as in Synthesis of Intermediate [1-C], except that 3,6-bis(trifluoromethyl)-9H-carbazole was used instead of 9H-carbazole-3,6-dicarbonitrile.

(290) (4) Synthesis of Intermediate [13-D]

(291) Intermediate [13-D] (yield: 76%) was synthesized in substantially the same manner as in Synthesis of Intermediate [1-E], except that Intermediate [13-C] was used instead of Intermediate [1-C].

(292) (5) Synthesis of Compound 13

(293) Compound 13 (yield: 26%) was synthesized in substantially the same manner as in Synthesis of Compound 1, except that Intermediate [13-B] was used instead of Intermediate [1-B], and Intermediate [13-D] was used instead of Intermediate [1-E].

Synthesis Example 3: Synthesis of Compound 19

(294) ##STR00121##

(1) Synthesis of Intermediate [19-A]

(295) Intermediate [19-A] (yield: 49%) was synthesized in substantially the same manner as in Synthesis of Intermediate [1-C], except that 2,6-difluoropyridin-3-ol was used instead of 9H-carbazole-3,6-dicarbonitrile.

(296) (2) Synthesis of Intermediate [19-B]

(297) Intermediate [19-B] (yield: 72%) was synthesized in substantially the same manner as in Synthesis of Intermediate [1-E], except that Intermediate [19-A] was used instead of Intermediate [1-C].

(298) (3) Synthesis of Compound 19

(299) Compound 19 (yield: 23%) was synthesized in substantially the same manner as in Synthesis of Compound 1, except that Intermediate [19-B] was used instead of Intermediate [1-E].

Synthesis Example 4: Synthesis of Compound 25

(300) ##STR00122##

(1) Synthesis of Intermediate [25-A]

(301) 2,6-dibromopyridine (1.0 eq), (2,6-difluoropyridin-3-yl)boronic acid (1.1 eq), Pd(PPh₃)₄ (0.002 eq), and K₂CO₃ (1.0 eq) were dissolved in tetrahydrofuran/H₂O (2:1, 0.1 M) and stirred at a temperature of 70° C. for 12 hours. The reaction mixture was cooled to room temperature, and an organic layer was extracted therefrom three times using dichloromethane and water. The extracted organic layer was dried using magnesium sulfate and concentrated, and purified by column chromatography to synthesize Intermediate [25-A] (yield: 68%).

(302) (2) Synthesis of Intermediate [25-B]

(303) Intermediate [25-B] (yield: 76%) was synthesized in substantially the same manner as in Synthesis of Intermediate [1-E], except that Intermediate [25-A] was used instead of Intermediate [1-C].

(304) (3) Synthesis of Compound 25

(305) Compound 25 (yield: 27%) was synthesized in substantially the same manner as in Synthesis of Compound 1, except that Intermediate [25-B] was used instead of Intermediate [1-E].

(306) ¹H NMR and LC-MS of Compounds synthesized according to Synthesis Examples 1 to 4 are shown in Table 1.

(307) Synthesis methods of compounds other than Compounds shown in Table 1 may also be easily recognized by those of ordinary skill in the art by referring to the synthesis mechanisms and source materials described herein above.

(308) TABLE-US-00001 TABLE 1 Compound LC-MS No. ¹H NMR (CDCl₃, 400 MHz)

found cal. 1 8.22-7.02 (m, 7H), 6.52-6.48 (m, 4H) 5.21 (d, 2H), 897.2 897.1 5.06 (d, 2H), 2.69 (s, 6H) 13 9.11 (s, 1H), 8.34-7.03 (m, 6H), 6.64 (s, 2H), 6.51-6.49 1025.9 1025.8 (m, 2H), 5.24 (d, 2H), 5.01 (d, 2H), 2.67 (s, 6H) 19 7.65-7.45 (m, 2H), 6.54-6.47 (m, 3H), 5.54 (d, 2H), 811.1 811.1 5.23 (d, 2H), 5.04 (d, 2H), 2.69 (s, 6H) 25 7.86-7.00 (m, 3H), 6.65 (d, 1H), 6.52-6.48 (m, 3H), 795.0 795.1 5.24 (d, 2H), 5.02 (d, 2H), 2.71 (s, 6H)

Evaluation Example

(309) Wavelengths of maximum emission ($\lambda_{\text{sub.max}}$) of Compounds synthesized according to Synthesis Examples 1 to 4 were measured, and sup.3MC state energy of Compounds synthesized according to Synthesis Examples 1 to 4 were obtained through quantum simulation. Results thereof are shown in Table 2. The wavelength of maximum emission ($\lambda_{\text{sub.max}}$) was measured by using a photoluminescence (PL) measurement apparatus. The sup.3MC state energy was calculated through quantum simulation by using a density functional theory (DFT) method and the Gaussian program (available from Gaussian Inc.) where the respective compound was structurally optimized at a level of B3LYP/6-31G(d,p) (e.g., using the B3LYP hybrid functional and the 6-31G(d,p) basis set).

(310) TABLE-US-00002 TABLE 2 sup.3MC state λ_{max} (nm) (eV) Comparative Compound 472 0.24 1 Comparative Compound 498 0.15 2 Comparative Compound 502 0.21 3 Comparative Compound 470 0.02 4 Compound 1 468 0.24 Compound 13 464 0.23 Compound 19 449 0.19 Compound 25 456 0.21 Comparative Compound 1  Comparative Compound 2  Comparative Compound 3  Comparative Compound 4 

(311) While the present application is not limited by any particular mechanism or theory, it is believed that as a bond between a ligand and a metal center becomes stronger, energy at which the sup.3MC state is formed in a triplet state further increases.

(312) From the results of Table 2, it can be seen that Compounds 1, 13, 19, and 25 according to one or more embodiments have high sup.3MC state energy, as compared with Comparative Example 4 including the bidentate ligand.

EXAMPLES

Example 1

(313) As an anode, a Corning 15 Ω/cm . sup.2 (1,200 Å) ITO glass substrate was cut to a size of 50 mm×50 mm×0.7 mm, sonicated with isopropyl alcohol and pure water each for 5 minutes, and then cleaned by exposure to ultraviolet rays and ozone for 30 minutes. Then, the ITO glass substrate was provided to a vacuum deposition apparatus. 2-TNATA was vacuum-deposited on the ITO glass substrate to form a hole injection layer having a thickness of 600 Å, and 4,4'-bis[N-(1-naphthyl)-N-phenyl aminobiphenyl (NPB) was vacuum-deposited on the hole injection layer to form a hole transport layer having a thickness of 300 Å. Compound 1 (dopant) (weight ratio of 10%) and 3,3-di(9H-carbazol-9-yl)biphenyl (mCBP) (host) were co-deposited on the hole transport layer to form an emission layer having a thickness of 300 Å. Then, diphenyl(4-(triphenylsilyl)phenyl)-phosphine oxide (TSPO1) was vacuum-deposited on the emission layer to form a hole blocking layer having a thickness of 50 Å. Then, Alq. sub.3 was deposited on the hole blocking layer to form an electron transport layer having a thickness of 300 Å, and an alkali metal halide LiF was deposited on the electron transport layer to form an electron injection layer having a thickness of 10 Å, and Al was vacuum-deposited to a thickness of 3,000 Å (cathode electrode) to form a LiF/Al electrode, thereby completing the manufacture of an organic light-emitting device.

Examples 2 to 4

(314) Organic light-emitting devices were manufactured in substantially the same manner as in Example 1, except that Compounds shown in Table 3 were each used instead of Compound 1 as a dopant in forming an emission layer.

Comparative Examples 1 to 4

(315) Organic light-emitting devices were manufactured in substantially the same manner as in

Example 1, except that Comparative Compounds 1 to 4 were respectively used instead of Compound 1 as a dopant in forming an emission layer.

(316) ##STR00127##

(317) The driving voltage, luminance, luminescent efficiency, and maximum emission wavelength of the organic light-emitting devices manufactured according to Examples 1 to 4 and Comparative Examples 1 to 4 were measured at a current density of 50 mA/cm.^{sup.2} by using a Keithley SMU 236 and a luminance meter PR650, and an amount of time (half lifespan) that lapsed when luminance was 50% of initial luminance. Results thereof are shown in Table 3.

(318) TABLE-US-00003 TABLE 3 Maximum Diving Current emission Half voltage density Luminance Efficiency wavelength lifespan Emission layer (V) (mA/cm.^{sup.2}) (cd/m.^{sup.2}) (cd/A) (nm) (hr) Example 1 Compound 1 5.32 50 4125 8.25 469 13 Example 2 Compound 13 5.42 50 4025 8.05 467 15 Example 3 Compound 19 5.48 50 4033 8.07 452 10 Example 4 Compound 25 5.46 50 4024 8.05 457 11 Comparative Comparative 6.24 50 3785 7.57 475 12 Example 1 Compound 1 Comparative Comparative 5.71 50 3570 7.14 501 14 Example 2 Compound 2 Comparative Comparative 6.85 50 3860 7.72 502 9 Example 3 Compound 3 Comparative Comparative 6.56 50 3870 7.74 470 3 Example 4 Compound 4

(319) From Table 3, it can be seen that the organic light-emitting devices of Examples 1 to 4, in which the compounds according to one or more embodiments are used as the dopant of the emission layer have a low driving voltage, high efficiency, and high color purity, as compared with the organic light-emitting devices of Comparative Examples 1 to 4. In addition, it can be seen that the lifespan is remarkably improved, as compared with the organic light-emitting devices of Comparative Examples 3 and 4.

(320) When the compounds according to one or more embodiments are used for the organic light-emitting device, high color purity may be implemented and excellent effects may be exhibited in terms of driving voltage, efficiency, and lifespan.

(321) The organic light-emitting device including the organometallic compound may have a low driving voltage, high efficiency, and a long lifespan and facilitate emission wavelength adjustment, thereby exhibiting high color purity.

(322) It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should be considered as available for other similar features or aspects in other embodiments.

(323) It will be understood that, although the terms “first,” “second,” “third,” etc., may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, a first element, component, region, layer or section described below could be termed a second element, component, region, layer or section, without departing from the spirit and scope of the present disclosure.

(324) Spatially relative terms, such as “beneath,” “below,” “lower,” “under,” “above,” “upper,” and the like, may be used herein for ease of explanation to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or in operation, in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” or “under” other elements or features would then be oriented “above” the other elements or features. Thus, the example terms “below” and “under” can encompass both an orientation of above and below.

(325) The device may be otherwise oriented (e.g., rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein should be interpreted accordingly.

(326) It will be understood that when an element or layer is referred to as being “on,” “connected

to,” or “coupled to” another element or layer, it can be directly on, connected to, or coupled to the other element or layer, or one or more intervening elements or layers may be present. In addition, it will also be understood that when an element or layer is referred to as being “between” two elements or layers, it can be the only element or layer between the two elements or layers, or one or more intervening elements or layers may also be present.

(327) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a” and “an” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes,” and “including,” when used in this specification, specify the presence of the stated features, integers, acts, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, acts, operations, elements, components, and/or groups thereof.

(328) As used herein, the terms “substantially,” “about,” and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent deviations in measured or calculated values that would be recognized by those of ordinary skill in the art.

Further, the use of “may” when describing embodiments of the present disclosure refers to “one or more embodiments of the present disclosure.” As used herein, the terms “use,” “using,” and “used” may be considered synonymous with the terms “utilize,” “utilizing,” and “utilized,” respectively. Also, the term “exemplary” is intended to refer to an example or illustration.

(329) Also, any numerical range recited herein is intended to include all subranges of the same numerical precision subsumed within the recited range. For example, a range of “1.0 to 10.0” is intended to include all subranges between (and including) the recited minimum value of 1.0 and the recited maximum value of 10.0, that is, having a minimum value equal to or greater than 1.0 and a maximum value equal to or less than 10.0, such as, for example, 2.4 to 7.6. Any maximum numerical limitation recited herein is intended to include all lower numerical limitations subsumed therein, and any minimum numerical limitation recited in this specification is intended to include all higher numerical limitations subsumed therein. Accordingly, Applicant reserves the right to amend this specification, including the claims, to expressly recite any sub-range subsumed within the ranges expressly recited herein.

(330) While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present disclosure as defined by the appended claims, and equivalents thereof.

Claims

1. An organometallic compound represented by Formula 1:

ML.sub.AL.sub.B, Formula 1 wherein, in Formula 1, M is selected from platinum (Pt), palladium (Pd), copper (Cu), zinc (Zn), silver (Ag), gold (Au), iridium (Ir), rhodium (Rh), cobalt (Co), ruthenium (Ru), rhenium (Re), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), and thulium (Tm), L.sub.A is a tridentate ligand represented by Formula 1A, and L.sub.B is a tridentate ligand represented by Formula 1B: ##STR00128## wherein, in Formulae 1A and 1B, ring A.sub.1 is a 5-membered ring including Y.sub.1, a C.sub.5-C.sub.60 carbocyclic ring in which a 5-membered ring including Y.sub.1 is condensed, or a C.sub.1-C.sub.60 heterocyclic ring in which a 5-membered ring including Y.sub.1 is condensed, ring A.sub.2 to ring A.sub.6 are each independently selected from a C.sub.5-C.sub.60 carbocyclic ring and a C.sub.1-C.sub.60 heterocyclic ring, Y.sub.1 to Y.sub.6 are each independently N or C, L.sub.1 is selected from *—C≡C—*, *—B(R.sub.7)—*, *—P(R.sub.7)—*, *—P(R.sub.7)(R.sub.8)—*, and *—Ge(R.sub.7)(R.sub.8)—*, L.sub.2 is selected from *—O—*, *—S—*, *—C(R.sub.7)(R.sub.8)—*, *—C(R.sub.7)=*, *=C(R.sub.7)—*, *—C(R.sub.7)=C(R.sub.8)—*,

---C(=O)---^* , ---C(=S)---^* , ---C=---^* , $\text{---B(R.sub.7)---}^*$, $\text{---N(R.sub.7)---}^*$, $\text{---P(R.sub.7)---}^*$, $\text{---Si(R.sub.7)(R.sub.8)---}^*$, $\text{---P(R.sub.7)(R.sub.8)---}^*$, and $\text{---Ge(R.sub.7)(R.sub.8)---}^*$, n1 is selected from 1, 2, and 3, and n2 is selected from 0, 1, 2, and 3, R.sub.1 to R.sub.6 and R.sub.8 are each independently selected from hydrogen, deuterium, ---F , ---Cl , ---Br , ---I , a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, $\text{---Si(Q.sub.1)(Q.sub.2)(Q.sub.3)}$, $\text{---N(Q.sub.1)(Q.sub.2)}$, $\text{---B(Q.sub.1)(Q.sub.2)}$, ---C(=O)(Q.sub.1) , $\text{---S(=O).sub.2(Q.sub.1)}$, and $\text{---P(=O)(Q.sub.1)(Q.sub.2)}$, R.sub.7 is selected from an amidino group, a hydrazino group, a hydrazono group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, $\text{---Si(Q.sub.1)(Q.sub.2)(Q.sub.3)}$, $\text{---N(Q.sub.1)(Q.sub.2)}$, $\text{---B(Q.sub.1)(Q.sub.2)}$, ---C(=O)(Q.sub.1) , $\text{---S(=O).sub.2(Q.sub.1)}$, and $\text{---P(=O)(Q.sub.1)(Q.sub.2)}$, two neighboring R.sub.1(s) among R.sub.1(s) in the number of b1, two neighboring R.sub.2(s) among R.sub.2(s) in the number of b2, two neighboring R.sub.3(s) among R.sub.3(s) in the number of b3, two neighboring R.sub.4(s) among R.sub.4(s) in the number of b4, two neighboring R.sub.5(s) among R.sub.5(s) in the number of b5, two neighboring R.sub.6(s) among R.sub.6(s) in the number of b6, and at least one of R.sub.1 and R.sub.7, R.sub.2 and R.sub.7, and Rs and R.sub.7 are optionally linked to form a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, b1 to b6 are each independently an integer from 1 to 10, * indicates a binding site to M, at least one substituent of the substituted C.sub.5-C.sub.60 carbocyclic group, the substituted C.sub.1-C.sub.60 heterocyclic group, the substituted C.sub.1-C.sub.60 alkyl group, the substituted C.sub.2-C.sub.60 alkenyl group, the substituted C.sub.2-C.sub.60 alkynyl group, the substituted C.sub.1-C.sub.60 alkoxy group, the substituted C.sub.3-C.sub.10 cycloalkyl group, the substituted C.sub.1-C.sub.10 heterocycloalkyl group, the substituted C.sub.3-C.sub.10 cycloalkenyl group, the substituted C.sub.1-C.sub.10 heterocycloalkenyl group, the substituted C.sub.6-C.sub.60 aryl group, the substituted C.sub.6-C.sub.60 aryloxy group, the substituted C.sub.6-C.sub.60 arylthio group, the substituted C.sub.1-C.sub.60 heteroaryl group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group is selected from: deuterium, ---F , ---Cl , ---Br , ---I , a hydroxyl group, a cyano group, a nitro group, an amidino group,

a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, and a C.sub.1-C.sub.60 alkoxy group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, and a C.sub.1-C.sub.60 alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(=O)(Q.sub.11), —S(=O).sub.2(Q.sub.11), and —P(=O)(Q.sub.11)(Q.sub.12); a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group; a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(=O)(Q.sub.21), —S(=O).sub.2(Q.sub.21), and —P(=O)(Q.sub.21)(Q.sub.22); and —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 are each independently selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a biphenyl group, and a terphenyl group, wherein, when ring A.sub.3 is selected from a C.sub.5-C.sub.60 carbocyclic ring and a C.sub.1-C.sub.60 heterocyclic ring other than a carbazole group, R.sub.5 is selected from deuterium, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60

aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(=O)(Q.sub.1), —S(=O).sub.2(Q.sub.1), and —P(=O)(Q.sub.1)(Q.sub.2).

2. The organometallic compound of claim 1, wherein, a group represented by ##STR00129## in Formula 1A is selected from groups represented by Formulae A1-1 to A1-7: ##STR00130## wherein, in Formulae A1-1 to A1-7, X.sub.11 is C(R.sub.11) or N, X.sub.12 is C(R.sub.12) or N, X.sub.13 is C(R.sub.13) or N, X.sub.14 is C(R.sub.14) or N, X.sub.15 is C(R.sub.15) or N, X.sub.16 is C(R.sub.16) or N, and X.sub.17 is C(R.sub.17) or N, X.sub.18 is C(R.sub.18)(R.sub.19), Si(R.sub.18)(R.sub.19), N(C.sub.18), P(C.sub.18), O, or S, R.sub.11 to R.sub.19 are each independently the same as defined in connection with R.sub.1 in Formula 1A, and * indicates a binding site to M, and *' indicates a binding site to (L.sub.1).sub.a1.

3. The organometallic compound of claim 1, wherein, ring A.sub.3, ring A.sub.4, and ring A.sub.6 are each independently selected from groups represented by Formulae 2-1(1) to 2-1(27), and ring A.sub.2 and ring A.sub.5 are each independently selected from groups represented by Formulae 2-2(1) to 2-2(19): ##STR00131## ##STR00132## ##STR00133## ##STR00134## ##STR00135## ##STR00136## ##STR00137## ##STR00138## wherein, in Formulae 2-1(1) to 2-1(27) and 2-2(1) to 2-2(19), Y.sub.15 is N or C bonded to M, X.sub.21 is N or C(R.sub.21), X.sub.22 is N or C(R.sub.22), X.sub.23 is N or C(R.sub.23), X.sub.24 is N or C(R.sub.24), X.sub.25 is N or C(R.sub.25), X.sub.26 is N or C(R.sub.26), X.sub.27 is N or C(R.sub.27), and X.sub.28 is N or C(R.sub.28), X.sub.29 is C(R.sub.29a)(R.sub.29b), Si(R.sub.29a)(R.sub.29b), N(R.sub.29), O, or S, X.sub.30 is C(R.sub.30a)(R.sub.30b), Si(R.sub.30a)(R.sub.30b), N(R.sub.30), O, or S, Z.sub.21 and Z.sub.22 are each independently N or P, R.sub.21 to R.sub.30, R.sub.25a to R.sub.30a and R.sub.25b to R.sub.30b are each independently the same as defined in connection with R.sub.2 to R.sub.6 in Formula 1, * indicates a binding site to M, and *' and *'' each indicate a binding site to a neighboring atom.

4. The organometallic compound of claim 3, wherein, ring A.sub.2 and ring A.sub.6 are each independently a group represented by Formula 2-2(1), ring A.sub.3 is a group represented by Formula 2-1(1), and ring A.sub.4 and ring A.sub.6 are each independently a group represented by Formula 2-1(17) or 2-1(26).

5. The organometallic compound of claim 1, wherein, Y.sub.1 and Y.sub.2 are each N, Y.sub.3 is C, a bond between Y.sub.1 and M is a covalent bond, a bond between Y.sub.2 and M is a coordinate bond, and a bond between Y.sub.3 and M is a covalent bond.

6. The organometallic compound of claim 1, wherein, n1 is 1 and n2 is 0; or n1 is 1 and n2 is 1.

7. The organometallic compound of claim 1, wherein, i) n1 is 1, n2 is 0, and L.sub.1 is *—B(R.sub.7)—*'; or ii) n1 is 1, n2 is 1, and L.sub.2 is *—O—*', L.sub.1 is *—B(R.sub.7)—*' and L.sub.2 is *—O—*', L.sub.2 is *—N(R.sub.7)—*', or L.sub.1 is *—B(R.sub.7)—*' and L.sub.2 is *—N(R.sub.7)—*'. .

8. The organometallic compound of claim 1, wherein, L.sub.A in Formula 1 is a group represented by Formula 1C: ##STR00139## wherein, in Formula 1C, ring A.sub.1, ring A.sub.2, Y.sub.1 to Y.sub.3, L.sub.1, n1, R.sub.1, R.sub.2, b1, and b2 are each independently the same as described in claim 1, X.sub.31 is C(R.sub.31) or N, X.sub.32 is C(R.sub.32) or N, X.sub.33 is C(R.sub.33) or N, X.sub.34 is C(R.sub.34) or N, X.sub.35 is C(R.sub.35) or N, X.sub.36 is C(R.sub.36) or N, and X.sub.37 is C(R.sub.37) or N, R.sub.31 to R.sub.37 are each independently the same as defined in connection with R.sub.3 in Formula 1A, and * indicates a binding site to M.

9. The organometallic compound of claim 1, wherein, in Formula 1B, Y.sub.4 and Y.sub.6 are each independently C or N, and a bond between Y.sub.4 and M and a bond between Y.sub.6 and M are each a coordinate bond.

10. The organometallic compound of claim 1, wherein L.sub.B in Formula 1 is a group represented by Formula 1B-11 or 1B-12: ##STR00140## wherein, in Formulae 1B-11 and 1B-12, Y.sub.5, ring A.sub.5, R.sub.5, and b5 are each independently the same as defined in claim 1, Z.sub.41 to Z.sub.44 are each independently N or P, X.sub.41 is C(R.sub.41) or N, X.sub.42 is C(R.sub.42) or N, X.sub.43 is C(R.sub.43) or N, and X.sub.44 is C(R.sub.44) or N, R.sub.41 to R.sub.46 are each independently the same as defined in connection with R.sub.4 and R.sub.6 in Formula 1B, and * indicates a binding site to M.
11. The organometallic compound of claim 10, wherein, Z.sub.41 to Z.sub.44 are each N.
12. The organometallic compound of claim 1, wherein, at least one selected from R.sub.1(s) in the number of b1, R.sub.2(s) in the number of b2, R.sub.3(s) in the number of b3, R.sub.4(s) in the number of b4, R.sub.5(s) in the number of b5, and R.sub.6(s) in the number of b6 is selected from a cyano group, a C.sub.1-C.sub.10 alkyl group substituted with at least one —F, a C.sub.1-C.sub.10 alkyl group substituted with at least one —Cl, a C.sub.1-C.sub.10 alkyl group substituted with at least one —Br, and a C.sub.1-C.sub.10 alkyl group substituted with at least one —I.
13. An organometallic compound being selected from Compounds 5, 11, 17, 23, 29, 35, 41, 47, 53, 59, and 65: ##STR00141## ##STR00142## ##STR00143##
14. An organic light-emitting device comprising: a first electrode; a second electrode facing the first electrode; an organic layer between the first electrode and the second electrode and comprising an emission layer; and at least one of the organometallic compound of claim 1.
15. The organic light-emitting device of claim 14, wherein, the organic layer comprises the organometallic compound.
16. The organic light-emitting device of claim 14, wherein, the emission layer comprises the organometallic compound.
17. The organic light-emitting device of claim 16, wherein, the emission layer further comprises a host, and an amount of the organometallic compound comprised in the emission layer is in a range of about 0.01 parts by weight to about 30 parts by weight based on 100 parts by weight of the emission layer.
18. The organic light-emitting device of claim 16, wherein, the emission layer emits a light having a wavelength of maximum emission of 450 nm or more and 550 nm or less.
19. The organic light-emitting device of claim 14, wherein, the first electrode is an anode, the second electrode is a cathode, the organic layer further comprises a hole transport region between the first electrode and the emission layer and an electron transport region between the emission layer and the second electrode, the hole transport region comprises a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof, and the electron transport region comprises a buffer layer, a hole blocking layer, an electron transport layer, an electron injection layer, or any combination thereof.
20. The organic light-emitting device of claim 19, wherein, the hole transport region comprises a p-dopant having a lowest unoccupied molecular orbital (LUMO) energy level of -3.5 eV or less.
21. An organometallic compound represented by Formula 1:
ML.sub.AL.sub.B, Formula 1 wherein, in Formula 1, M is selected from platinum (Pt), palladium (Pd), copper (Cu), zinc (Zn), silver (Ag), gold (Au), iridium (Ir), rhodium (Rh), cobalt (Co), ruthenium (Ru), rhenium (Re), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), and thulium (Tm), L.sub.A is a tridentate ligand represented by Formula 1A, and L.sub.B is a tridentate ligand represented by Formula 1B: ##STR00144## wherein, in Formulae 1A and 1B, ring A.sub.1 is a 5-membered ring including Y.sub.1, a C.sub.5-C.sub.60 carbocyclic ring in which a 5-membered ring including Y.sub.1 is condensed, or a C.sub.1-C.sub.60 heterocyclic ring in which a 5-membered ring including Y.sub.1 is condensed, ring A.sub.2 to ring A.sub.6 are each independently selected from a C.sub.5-C.sub.60 carbocyclic ring and a C.sub.1-C.sub.60 heterocyclic ring, Y.sub.1 to Y.sub.6 are each independently N or C, L.sub.1 is selected from *—C≡C—*, *—B(R.sub.7)—*, *—P(R.sub.7)—*, *—P(R.sub.7)

(R.sub.8)—*', and *—Ge(R.sub.7)(R.sub.8)—*', L.sub.2 is selected from *—O—*', *—S—*', *—C(R.sub.7)(R.sub.8)—*', *—C(R.sub.7)=*', *=C(R.sub.7)—*', *—C(R.sub.7)=C(R.sub.8)—*', *—C(=O)—*', *—C(=S)—*', *—C=C—*', *—B(R.sub.7)—*', *—N(R.sub.7)—*', *—P(R.sub.7)—*', *—Si(R.sub.7)(R.sub.8)—*', *—P(R.sub.7)(R.sub.8)—*', and *—Ge(R.sub.7)(R.sub.8)—*', R.sub.1 to R.sub.6 and Rs are each independently selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(=O)(Q.sub.1), —S(=O).sub.2(Q.sub.1), and —P(=O)(Q.sub.1)(Q.sub.2), R.sub.7 is selected from an amidino group, a hydrazino group, a hydrazono group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C.sub.1-C.sub.60 alkyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkenyl group, a substituted or unsubstituted C.sub.2-C.sub.60 alkynyl group, a substituted or unsubstituted C.sub.1-C.sub.60 alkoxy group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkyl group, a substituted or unsubstituted C.sub.3-C.sub.10 cycloalkenyl group, a substituted or unsubstituted C.sub.1-C.sub.10 heterocycloalkenyl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryl group, a substituted or unsubstituted C.sub.6-C.sub.60 aryloxy group, a substituted or unsubstituted C.sub.6-C.sub.60 arylthio group, a substituted or unsubstituted C.sub.1-C.sub.60 heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(=O)(Q.sub.1), —S(=O).sub.2(Q.sub.1), and —P(=O)(Q.sub.1)(Q.sub.2), two neighboring R.sub.1(s) among R.sub.1(s) in the number of b1, two neighboring R.sub.2(s) among R.sub.2(s) in the number of b2, two neighboring R.sub.3(s) among R.sub.3(s) in the number of b3, two neighboring R.sub.4(s) among R.sub.4(s) in the number of b4, two neighboring R.sub.5(s) among R.sub.5(s) in the number of b5, two neighboring R.sub.6(s) among R.sub.6(s) in the number of b6, and at least one of R.sub.1 and R.sub.7, R.sub.2 and R.sub.7, and R.sub.3 and R.sub.7 are optionally linked to form a substituted or unsubstituted C.sub.5-C.sub.60 carbocyclic group or a substituted or unsubstituted C.sub.1-C.sub.60 heterocyclic group, b1 to b6 are each independently an integer from 1 to 10, * indicates a binding site to M, at least one substituent of the substituted C.sub.5-C.sub.60 carbocyclic group, the substituted C.sub.1-C.sub.60 heterocyclic group, the substituted C.sub.1-C.sub.60 alkyl group, the substituted C.sub.2-C.sub.60 alkenyl group, the substituted C.sub.2-C.sub.60 alkynyl group, the substituted C.sub.1-C.sub.60 alkoxy group, the substituted C.sub.3-C.sub.10 cycloalkyl group, the substituted C.sub.1-C.sub.10 heterocycloalkyl group, the substituted C.sub.3-C.sub.10 cycloalkenyl group, the substituted C.sub.1-C.sub.10 heterocycloalkenyl group, the substituted C.sub.6-C.sub.60 aryl group, the substituted C.sub.6-C.sub.60 aryloxy group, the substituted C.sub.6-C.sub.60 arylthio group, the substituted C.sub.1-C.sub.60 heteroaryl group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed

heteropolycyclic group is selected from: deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, and a C.sub.1-Coo alkoxy group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, and a C.sub.1-C.sub.60 alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-Coo heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(=O)(Q.sub.11), —S(=O).sub.2(Q.sub.11), and —P(=O)(Q.sub.11)(Q.sub.12); a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group; a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(=O)(Q.sub.21), —S(=O).sub.2(Q.sub.21), and —P(=O)(Q.sub.21)(Q.sub.22); and —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), and —P(=O)(Q.sub.31)(Q.sub.32), and Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 are each independently selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-Coo alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a biphenyl group, and a terphenyl group, n1 is selected from 1, 2, and 3, and n2 is selected from 0, 1, 2, and 3, provided that when n2 is 0, L.sub.A in Formula 1 is a group represented by Formula 1C: ##STR00145## wherein, in Formula 1C, ring A.sub.1, ring A.sub.2, Y.sub.1 to Y.sub.3, L.sub.1, n1, R.sub.1, R.sub.2, b1, and b2 are each independently the same as described in Formula 1A, X.sub.31 is C(R.sub.31) or N, X.sub.32 is C(R.sub.32) or N, X.sub.33 is C(R.sub.33) or N, X.sub.34 is C(R.sub.34) or N, X.sub.35 is C(R.sub.35) or N, X.sub.36 is C(R.sub.36) or N, and X.sub.37 is C(R.sub.37) or N, R.sub.31 to R.sub.37 are each independently the same as defined in connection with R.sub.3 in Formula 1A, and * indicates a binding site to M.

